

**KWAME NKRUMAH UNIVERSITY OF SCIENCE AND  
TECHNOLOGY, KUMASI**



**HOSPITAL CHATBOT FOR COLOR-CODED CLINICAL  
PATHWAYS**

**USING BIDIRECTIONAL ENCODER REPRESENTATIONS FROM  
TRANSFORMERS**

BY

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## DECLARATION

We hereby declare that this submission is our own work towards the award of the Bachelor of Science degree in Mathematics and that, to the best of our knowledge, it contains no material previously published by another person nor material which has been accepted for the award of any other degree of the university, except where due acknowledgment has been made in the text.

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## **DEDICATION**

To my family, lecturers, and the patients whose care motivates this work.

# ABSTRACT

**Triage** is the structured process of assigning each arriving patient an urgency level so that hospital resources match clinical need. In low- and middle-income countries, including Ghana, overcrowding, delayed first assessment, and *acuity discontinuity* (urgency colour stops guiding care after intake) affect outpatient departments, wards, and emergency streams alike. This project develops a hospital chatbot for color-coded clinical pathways using Bidirectional Encoder Representations from Transformers (BERT): it maps unstructured *chief complaints* (unstructured descriptions of why the patient has come) to South African Triage Scale (SATS) colours and clinical pathways. The mathematical model converts English patient text to a colour through a fine-tuned Biomedical Bidirectional Encoder Representations from Transformers (BioBERT) classifier. Tokenisation  $\tau$ , three embedding components, multi-head self-attention, and Softmax classification (framed as a Categorical distribution over five mutually exclusive acuity levels) are specified with full formulas. A fixed map  $f_{\text{SATS}}$  converts predicted acuity levels to Red, Orange, Yellow, or Green. The Triage Early Warning Score (TEWS) is presented as the parallel hospital standard for objective vital signs; when vitals are unavailable at intake, the text model supplies the initial colour. A clinical relevance gate filters non-medical input before classification. Pathway cards  $P(C)$  bind each colour to destination, maximum waiting time, and escalation at KNUST University Hospital.

Holdout evaluation on the 80,000-pair Triagegeist benchmark reports 99.92% accuracy and approximately 100 ms median inference latency. Synthetic Minority Over-sampling Technique (SMOTE) and stratified training mitigate class imbalance, raising Level 1 (Red) recall from 98.14% to 100% at a small precision cost, prioritising a safety-first clinical trade-off. The chatbot is positioned as nurse-overridable triage assist, not as a replacement for clinical judgment.

The methodology is illustrated on a running complaint at KNUST University Hospital: “*I have a headache and feel feverish. My body aches and I feel weak.*” System architecture and the prototype `/predict` application programming interface (API) are documented in Chapter 3 and Appendix 5.4. Results and conclusion are reported in later chapters.

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## ABBREVIATIONS

**SATS** South African Triage Scale

**TEWS** Triage Early Warning Score

**OPD** Outpatient Department

**LMIC** Low- and Middle-Income Country

**KATH** Komfo Anokye Teaching Hospital

**KNUST** Kwame Nkrumah University of Science and Technology

**NLP** Natural Language Processing

**BERT** Bidirectional Encoder Representations from Transformers

**BioBERT** Biomedical Bidirectional Encoder Representations from Transformers

**MLM** Masked Language Modelling

**FFN** Feed-Forward Network

**AVPU** Alert, Voice, Pain, Unresponsive (consciousness scale)

**API** Application Programming Interface

**SMOTE** Synthetic Minority Over-sampling Technique

**NLL** Negative Log-Likelihood

**HIS** Hospital Information System

**HIT** Hospital Information Technology

**HR** Heart Rate

**RR** Respiratory Rate

**CLS** Classification token ( [CLS] ) used as the sequence summary in BERT

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# CHAPTER 1

## INTRODUCTION

### 1.1 Background

Hospitals in Ghana operate under sustained pressure: crowded waiting areas, limited nursing staff, and long queues before a clinician sees a new arrival. Triage is the mechanism that aligns patient urgency with resource allocation. In well-resourced systems, urgency is reassessed as patients move through diagnostics, wards, and outpatient streams. In resource-constrained settings, however, standardised protocols such as the South African Triage Scale (SATS) are often applied only at the first desk. Once a patient leaves that point, their colour-coded acuity may no longer steer bed allocation, laboratory orders, or re-assessment timers. We call this failure *acuity discontinuity*.

Teaching hospitals in Ghana, including KNUST University Hospital, apply SATS to structure intake across clinical departments, not only in a single emergency bay. Colour codes (Red, Orange, Yellow, Green, and Blue for dignity protocols) express how quickly a patient must be seen and where they should stream next: resuscitation, acute beds, urgent waiting, or routine outpatient care. Nurses at triage also compute a Triage Early Warning Score (TEWS) from objective vital signs. Yet many delays begin earlier, when a patient arrives with only words: a headache, fever, weakness, or abdominal pain described in free text before any thermometer or pulse oximeter reading is available.

This project proposes a digital *Flow Management Engine* in the form of a hospital chatbot for color-coded clinical pathways using Bidirectional Encoder Representations from Transformers (BERT). The chatbot accepts a single English free-text *chief complaint* (the primary reason the patient presents, in their own words; see Section 3.1.1) and returns a SATS colour reflecting urgency together with acuity probabilities and a clinical pathway card that can guide routing across outpatient departments, wards, and intake queues. The core contribution is mathematical: a pipeline from unstructured complaint text to acuity probabilities, colour, and pathway card, developed in Chapter 3. The deployed encoder is BioBERT, a biomedical BERT checkpoint fine-tuned for five-class triage.

### 1.2 Motivation and significance

Three operational facts motivate the work.

1. **Text arrives before vitals.** At home, in a queue, or at a kiosk, patients can describe symptoms long before six TEWS parameters are measured. A language model that produces an initial SATS colour therefore fills a real temporal gap between arrival and nurse assessment (Stewart et al., 2023).
2. **Colour must travel with the patient.** SATS colours already exist in Ghanaian hospital policy (Rominski et al., 2014; KATH Emergency Department, 2018). The unsolved problem is not inventing colours, but binding them to destination, maximum waiting time  $T_{\max}$ , and escalation so that urgency does not vanish after the intake desk (South African Triage Group, 2012; Sundström et al., 2014).
3. **Green congestion harms Red and Orange care.** Non-urgent presentations occupying acute capacity (sometimes called the Green Tag Burden) delay time-critical patients (Yilmaz, 2021). Digital diversion of Green pathways toward outpatient or specialty streams is therefore a flow-management requirement, not a convenience feature.

For mathematics students and hospital stakeholders alike, the significance is twofold. First, the thesis shows how contemporary Transformer mathematics (tokenisation, embeddings, self-attention, Softmax) can be stated rigorously enough for examination while remaining deployable as a chatbot API. Second, it links that mathematics to an auditable hospital protocol: every predicted acuity level maps through a fixed function  $f_{\text{SATS}}$  to a colour and then to a pathway card  $P(C)$  at KNUST University Hospital.

## 1.3 Problem statement

Low- and middle-income country (LMIC) hospitals struggle with crowded waiting rooms and limited human resources at intake. Patients may wait hours before first assessment. Critical cases are not always identified quickly, while less urgent presentations congest acute zones. Medical systems are confusing; without immediate direction, patients do not know whether to join the emergency stream, the polyclinic, or self-care advice.

Manual, paper-based triage creates acuity discontinuity: patient status becomes static rather than dynamic. Five systemic failures follow.

1. **Pre-triage bottleneck.** Delay accumulates before any colour is assigned, because the first human assessor is overloaded.
2. **Lost urgency.** Patients triaged Orange or Red can lose time-critical priority when they move to imaging, laboratories, or wards, because no digital record preserves their colour and countdown timer.

3. **Green congestion.** Non-urgent (Green) patients lack automated guidance toward outpatient or self-care routes, blocking beds and staff needed for life-saving care.
4. **Text-only arrival.** At the door, vitals are often incomplete; subjective language (“worst headache”, “feverish and weak”) is available but unused by TEWS until measurements exist.
5. **Protocol gap for chatbots.** Generic consumer symptom checkers are not aligned to SATS hospital pathways or local operational routing (for example outpatient urgent stream versus acute resuscitation).

Without a framework to manage flow hospital-wide, overcrowding and resource mismatch persist; the most vulnerable patients become invisible to the system after the first desk.

## 1.4 Aim and objectives

### 1.4.1 General objective

To design and specify a hospital chatbot that maps an unstructured English chief complaint to a SATS colour-coded clinical pathway, using a fine-tuned BioBERT classifier and protocol tables aligned to Ghanaian hospital intake.

### 1.4.2 Specific objectives

1. **Automate triage classification.** Develop a chatbot that classifies a single unstructured chief complaint submitted by the patient or attendant, producing acuity probabilities, a SATS colour, and a pathway card reflecting clinical urgency.
2. **Guide clinical pathways.** Provide immediate, actionable next-step guidance: critical (Red) cases toward urgent care; non-urgent (Green) cases toward self-care or routine clinic streams.
3. **Integrate with hospital protocols.** Align decision logic with SATS and evidence-based pathway tables so recommendations remain safe and auditable, with TEWS documented as the parallel nurse-facing vital-sign standard.
4. **Collect anonymised data for refinement.** Use de-identified intake records to improve the classification model and to inform administrators about patient flow and common presentations.

## 1.5 Research questions

The project is organised around the following research questions.

- RQ1** Can unstructured English chief complaints be mapped to SATS acuity levels 1–5 with clinically usable holdout accuracy?
- RQ2** Does correct acuity prediction imply correct colour assignment under the fixed map  $f_{\text{SATS}}$ ?
- RQ3** How should predicted colours translate into pathway cards  $P(C)$  (destination,  $T_{\text{max}}$ , actions, escalation) at KNUST University Hospital?
- RQ4** Can a chatbot deliver pre-desk acuity and pathway guidance from text alone before nurse vitals are complete?
- RQ5** How well does a clinical relevance gate reject non-clinical input before BioBERT runs?
- RQ6** How does class imbalance affect Level 1 (Red) recall, and what mitigation (including SMOTE) is appropriate?
- RQ7** What Softmax confidence behaviour should trigger mandatory nurse oversight at deployment?
- RQ8** Can Green-pathway diversion be specified so that low-acuity patients leave acute queues without removing nurse authority?

## 1.6 Scope and limitations

**In scope.** English chief complaint  $X$ ; clinical relevance gate; WordPiece tokenisation; BioBERT-base encoding ( $M = 128$ ,  $D = 768$ , twelve layers); Softmax classification over five mutually exclusive acuity levels; deterministic colour map  $f_{\text{SATS}}$ ; pathway cards  $P(C)$  for gate reject, Red, Orange, Yellow, and Green at KNUST University Hospital; TEWS mathematics as the documented parallel vital-sign standard; holdout evaluation on the Triagegeist English corpus; Curatio prototype exposing `/predict`.

**Out of scope.** Full Bayesian network structure learning; learned fusion of TEWS with NLP as a single trained head; prospective randomised clinical trial; full Hospital Information System (HIS) integration; multilingual (for example Twi) intake models; Blue dignity pathway prediction from the five-class head (Blue remains a protocol category on arrival, not a Softmax class).



The chatbot is a **triage assist**: nurses retain override authority, especially for low-confidence Softmax outputs and all vital-sign decisions (Stewart et al., 2023; South African Triage Group, 2012).

## 1.7 Contributions

- A fully specified mathematical pipeline from free-text complaint to SATS colour and pathway card, including tokenisation  $\tau$ , embeddings, multi-head self-attention, and Categorical Softmax classification.
- An evaluated BioBERT Stage 3 fine-tune on 80,000 Triagegeist English pairs, with baseline and SMOTE checkpoints, latency, and per-class metrics.
- Protocol-mapped pathway tables and department routing for seventeen units at KNUST University Hospital, including Green diversion.
- A deployable chatbot//predict prototype with medical gating, documented alongside TEWS as the nurse vital-sign path.

## 1.8 Organisation of the thesis

Chapter 2 reviews hospital overcrowding, SATS and TEWS, Transformer and BioBERT literature, and NLP for triage, ending in the research gap. Chapter 3 develops the methodology: chief complaints and data provenance, TEWS, a neural-network and Transformer primer, then the full text pipeline through Softmax, colour mapping, pathways, and the gate. Chapter 4 analyses the BioBERT model first, then reports holdout performance, pathway routing, and anticipated hospital impact. Chapter 5 summarises findings against the research questions, restates limitations, and outlines future work.

## CHAPTER 2

### LITERATURE REVIEW

#### 2.1 Introduction

This chapter situates the project in published work on hospital overcrowding, SATS and TEWS practice, clinical pathways, Transformer language models, and natural language processing (NLP) for triage. The review is organised thematically rather than as an annotated bibliography: each section synthesises what is known, what remains contested, and what implication follows for a Ghanaian hospital chatbot. The chapter closes with an explicit research gap that Chapter 3 addresses for the text path.

#### 2.2 Hospital overcrowding and triage delay

Overcrowding in low- and middle-income country (LMIC) hospitals affects outpatient departments, emergency intake, and inpatient wards simultaneously (KATH Emergency Department, 2018; Mitchell et al., 2023). Delayed triage lengthens time-to-provider, increases left-without-being-seen rates, and correlates with poor outcomes for time-sensitive conditions. Waiting time interacts with colour assignment: when reassessment is weak, deterioration after the first desk can go unnoticed (Nurchis et al., 2025).

A structural driver is competition for acute capacity by non-urgent patients: the Green Tag Burden (Yilmaz, 2021). In LMICs, understaffing and limited diagnostics intensify that competition. Digital diversion of Green patients to outpatient, minors, or specialty streams is therefore a flow-management requirement for colour-coded pathways, not merely a convenience feature (Sundström et al., 2014).

**Implication for this project.** Any chatbot that assigns SATS colours must also attach destinations and timers; otherwise colour remains a label that dies at the desk (acuity discontinuity in Chapter 1).

#### 2.3 SATS in LMIC and Ghanaian practice

The South African Triage Scale (SATS) provides a five-colour framework (Red, Orange, Yellow, Green, Blue) with maximum waiting times  $T_{\max}$  and destination rules (South African Triage Group, 2012). Implementation at Komfo Anokye Teaching Hospital (KATH) and related Ghanaian settings demonstrated that standardised colour coding

can structure flow even when staffing is constrained, provided urgency information travels with the patient (Rominski et al., 2014; KATH Emergency Department, 2018). SATS combines physiological early warning (TEWS) with clinical discriminators that can escalate colour when high-risk symptoms are present even if vitals are only mildly abnormal (for example central chest pain). Colour codes therefore already exist in hospital policy. The research gap is not inventing colours, but (i) assigning them from free-text complaints before full vitals, and (ii) binding colours to pathway protocols so acuity does not disappear after intake.

Teaching hospitals such as KNUST University Hospital apply similar colour logic across clinical departments, not only in a single emergency bay. That multi-department setting motivates pathway cards that name outpatient urgent streams, surgery, obstetrics, pediatrics, and support units rather than a single undifferentiated “ED” destination.

## 2.4 TEWS and vital-sign triage

The Triage Early Warning Score (TEWS) aggregates six SATS parameters (mobility, respiratory rate (RR), heart rate (HR), temperature, Alert–Voice–Pain–Unresponsive (AVPU) consciousness, and trauma flag) into a single integer  $T$  (South African Triage Group, 2012). Nurses use  $T$  at triage to assign a colour band: Red for  $T > 7$ , Orange for  $5 \leq T \leq 6$ , Yellow for  $3 \leq T \leq 4$ , Green for  $T \leq 2$ . TEWS is deterministic and objective; it does not read language.

Subjective symptoms such as “worst headache of my life” or “feverish and weak” may appear in the complaint before corresponding vitals exist. Chapter 3 therefore presents TEWS mathematics as the hospital standard *parallel* to the text model: when vitals are available, nurses compute  $T$ ; when they are not, the BioBERT path supplies an initial colour for routing assist. This thesis does not claim that Softmax replaces TEWS.

## 2.5 Clinical pathways as operational roadmaps

Clinical pathways are step-by-step care roadmaps: destination, diagnostics, time targets, and escalation (Sundström et al., 2014). In SATS terms, colour without a pathway is incomplete: Red must mean immediate resuscitation access; Orange must mean a short countdown; Green must mean diversion away from acute beds when clinically appropriate (South African Triage Group, 2012; Yilmaz, 2021).

Pathway literature supports treating colour as operational, not decorative. Under-triage (assigning too low an urgency) is a recognised quality trigger in emergency medicine (Berkowitz et al., 2019); over-triage wastes scarce acute capacity. A chatbot that emits pathway cards  $P(C)$  must therefore be evaluated not only on label accuracy but on whether routing rules are protocol-faithful and nurse-overridable.

## 2.6 Attention, BERT, and BioBERT

Sequential recurrent language models read text predominantly in one direction, limiting context for ambiguous clinical phrases. Vaswani et al. (2017) introduced scaled dot-product self-attention, allowing each token to attend to all others in parallel within a fixed window. That property matters for triage language, where negation and severity modifiers (“no pain”, “severe headache”) must reshape neighbouring words.

Devlin et al. (2019) built BERT (Bidirectional Encoder Representations from Transformers) on this idea: masked language modelling (MLM) pre-training yields contextual embeddings

$$E(t_i) = E_{\text{word}}(t_i) + E_{\text{pos}}(i) + E_{\text{seg}}(s)$$

used across downstream tasks. BERT is an encoder stack: it produces contextual vectors for classification and tagging rather than open-ended generation.

Lee et al. (2020) continued pre-training on PubMed and PubMed Central text to produce Biomedical BERT (BioBERT), a domain-specific encoder whose embedding space places clinical terms nearer related concepts. Fine-tuning BioBERT on nurse-labeled chief complaints yields a supervised classifier from text to acuity levels. Classification with a fixed five-class label set reduces open-ended generative hallucination risk compared with unconstrained large language models that invent diagnoses in free prose (Stewart et al., 2023).

## 2.7 NLP for emergency and hospital triage

Stewart et al. (2023) reviewed NLP at emergency triage: chief-complaint classification, syndromic surveillance, and decision support remain active research areas with heterogeneous datasets and uneven prospective validation. Gligorijević et al. (2018) showed that deep attention models can stratify emergency department patients from unstructured text. Class imbalance is a recurring theme: rare life-threatening presentations (Level 1 / Red) are under-represented relative to routine Yellow and Green cases, so accuracy alone can hide unsafe under-triage (Berkowitz et al., 2019; Chawla et al., 2002).

Most published systems target high-income corpora or electronic health record notes written by clinicians. Formal mathematics from *patient* free text to SATS colour for Ghanaian hospital intake, with pathway routing aligned to local department structure, is under-specified in the literature. Synthetic benchmarks such as Triagegeist (Laitinen-Fredriksson Foundation, 2025) make large-scale supervised fine-tuning possible, but they do not substitute for prospective local validation.

## 2.8 Conversational agents and digital triage tools

Symptom-assessment applications and ED prediction chatbots show that conversational NLP can support triage-like advice before the nurse desk (Stewart et al., 2023). Validation studies emphasise safety, under-triage monitoring, nurse override, and comparison with conventional triage. Consumer tools often return free-form advice that is not mapped to hospital colour protocols or local destinations.

Our chatbot is deliberately constrained: it outputs acuity probabilities, a SATS colour, and a pathway card aligned to SATS and KNUST University Hospital routing—not free-form diagnosis essays. A clinical relevance gate rejects non-medical chat so that BioBERT is not run on greetings or sports conversation. That constrained design follows the safety emphasis in the triage NLP review literature.

## 2.9 Synthesis: what prior work supplies and what it leaves open

Table 2.1 summarises the strands reviewed above.

| Strand                      | What exists                                                                                           | Open for this project                                                      |
|-----------------------------|-------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| Overcrowding / Green burden | Evidence that delay and non-urgent congestion harm flow (Mitchell et al., 2023; Yilmaz, 2021)         | Digital colour + diversion rules at Ghanaian intake                        |
| SATS / TEWS                 | Colour bands and vital-score algebra (South African Triage Group, 2012; Rominski et al., 2014)        | Text-first colour before vitals; TEWS kept parallel                        |
| Pathways                    | Operational roadmaps and diversion models (Sundström et al., 2014)                                    | Explicit $P(C)$ tables for seventeen departments                           |
| Transformers / BioBERT      | Contextual encoders for biomedical text (Vaswani et al., 2017; Devlin et al., 2019; Lee et al., 2020) | Examinable pipeline $X \rightarrow \hat{y} \rightarrow C \rightarrow P(C)$ |
| Triage NLP                  | ED text classifiers and reviews (Stewart et al., 2023; Gligorijević et al., 2018)                     | SATS-aligned, pathway-bound chatbot for LMIC intake                        |

Table 2.1: Literature synthesis: available foundations versus the gap this project fills.

## 2.10 Research gap

Existing work separates vital-sign scores (TEWS), manual SATS colour assignment, and NLP classifiers trained largely abroad. There is no unified, examinable

mathematical account of how a free-text English chief complaint at KATH-style or teaching-hospital intake becomes a SATS colour and pathway card using BioBERT, while TEWS remains documented as the nurse-facing vital standard and nurses retain override.

In short, prior art provides colours, scores, and growing NLP evidence; it does not provide the integrated text path

$$X \xrightarrow{\tau} H^{(0)} \xrightarrow{\text{BioBERT}} \hat{\mathbf{y}} \xrightarrow{f_{\text{SATS}}} C_{\text{NLP}} \rightarrow P(C_{\text{NLP}})$$

with clinical gating, Softmax framed as a Categorical distribution over mutually exclusive acuity levels, and department routing tables matched to KNUST University Hospital.

Three concrete absences define the gap this thesis fills:

1. **Examinable mathematics.** Published triage NLP papers often report accuracy without deriving tokenisation, attention Softmax, and classification Softmax in one consistent notation suitable for a mathematics project defence.
2. **Pathway binding.** Many systems stop at a predicted label; few attach SATS-aligned  $T_{\text{max}}$ , destination, and escalation for a named Ghanaian teaching-hospital department map.
3. **Parallel TEWS documentation.** Text classifiers are sometimes presented as if vitals were obsolete; this project keeps TEWS explicit as the nurse vital-sign standard while the chatbot covers the text-first moment.

Chapter 3 fills that gap for the text path; Chapter 4 reports evaluation.

## CHAPTER 3

### METHODOLOGY

This chapter specifies the mathematical pipeline from an arbitrary chief complaint to a South African Triage Scale (SATS) colour and clinical pathway. Where concrete numbers aid understanding, we refer to a recurring illustrative example at KNUST University Hospital: “*I have a headache and feel feverish. My body aches and I feel weak.*” The same steps apply to any valid clinical text that passes the relevance gate in Section 3.17.

The chatbot supports triage staff; it does not diagnose disease or replace clinical judgment.

#### 3.1 Methodological overview

The hospital chatbot draws on two complementary ideas from clinical practice; this thesis develops the text-based path in full and documents the Triage Early Warning Score (TEWS) as the parallel vital-sign standard.

First, **clinical pathways** encode hospital protocols: symptoms and context map to an urgency colour and a destination (resuscitation bay, acute bed, urgent waiting, outpatient stream). Pathways are the product the chatbot shows to patients and staff after classification.

Second, TEWS is the nurse-facing vital-sign score used at SATS triage desks. When heart rate (HR), respiratory rate (RR), mobility, temperature, Alert, Voice, Pain, Unresponsive (AVPU) consciousness, and trauma status are measured, TEWS produces a deterministic integer  $T$  and colour band  $C_{\text{TEWS}}(T)$ . TEWS is explained fully in Section 3.3 because it is the objective hospital standard; the chatbot’s primary engine for *free-text chief complaints* at intake is a deep attention model (Biomedical BERT, BioBERT), not a weighted sum of vitals.

When vitals are not yet available, the text model below supplies the intake colour. Nurses may later record vitals and apply TEWS independently; this thesis specifies the natural language processing (NLP) path only.

##### 3.1.1 Chief complaints

The *chief complaint* is the primary reason the patient presents, expressed in their own words (or an attendant’s) as unstructured free text at intake, before vitals, physical

examination, or formal diagnosis. Nurses and triage staff traditionally record this as prose on paper or electronic forms; it captures symptom onset, severity, and context (for example, “headache and feverish, body aches, feel weak”). It is not a diagnosis, not a TEWS vital-sign score, and not a full medical history: it is only the presenting problem text  $X$  that this thesis classifies.

The chatbot accepts one **English** chief-complaint string per submission. The running example “*I have a headache and feel feverish. My body aches and I feel weak.*” illustrates a typical Yellow-level presentation used throughout this chapter; pipeline symbol  $X$  appears in Table 3.1 and Figure 3.1.

### 3.1.2 Training data provenance

Model training and holdout evaluation draw on two distinct data layers.

**Training and evaluation corpus.** Stage 3 fine-tuning uses the *Triagegeist* synthetic emergency-department benchmark released by the Laitinen-Fredriksson Foundation on Kaggle (Laitinen-Fredriksson Foundation, 2025). Structured intake records (train.csv, acuity labels) are inner-joined with free-text rows (chief\_complaints.csv) on patient\_id, yielding  $N = 80,000$  English chief\_complaint\_raw / triage\_acuity pairs. Vitals and demographics in the source files are excluded so the NLP path matches single-shot text input at intake (Section 3.14.3). Five-level competition labels map to SATS colours via  $f_{\text{SATS}}$ . A stratified 90/10 split (approximately 72,000 train / 8,000 holdout, seed 42) supplies all reported accuracy and recall figures in Chapter 4. This corpus was **not** collected at KNUST University Hospital.

**KNUST-informed contextual examples.** Following a site visit to KNUST University Hospital, the authors generated supplementary English chief complaints based on observations of intake workflow, SATS colour assignment, and department structure. These examples test the prototype chatbot, validate pathway routing (Section 3.16), and illustrate the pipeline (for example, “*I have a headache and feel feverish. My body aches and I feel weak.*” in Table 3.21); this supports qualitative analysis only. They are **not** merged into the 80,000-pair fine-tuning corpus. Prospective collection of nurse-labeled complaints at KNUST University Hospital for model retraining remains future work (Chapter 5).

### 3.1.3 BERT and bidirectional context

Older language models read text sequentially (left-to-right or right-to-left). Vaswani et al. (2017) showed that *self-attention* lets every token attend to every other token in one pass. Devlin et al. (2019) built Bidirectional Encoder Representations from Transformers (BERT) on this mechanism: during pre-training, random tokens are



masked and predicted from full left *and* right context via masked language modelling (MLM). The result is contextual embeddings: the vector for “feverish” differs depending on neighbouring words such as “headache” or “not”.

### 3.1.4 BioBERT for biomedical text

General BERT embeddings under-represent clinical terms. Lee et al. (2020) continued MLM pre-training on PubMed and PMC, then released BioBERT with the same transformer skeleton as BERT but an embedding space tuned for biomedical language. For triage, BioBERT is fine-tuned on the Triagegeist corpus described in Section 3.1.2: supervised learning maps English complaint text  $X$  to acuity level probabilities  $\hat{y}$ , then to SATS colour via a fixed map  $f_{\text{SATS}}$ .

### 3.1.5 The [CLS] summary token

Classification does not read all token outputs separately. An artificial token [CLS] is inserted at position  $i = 0$ . After  $L = 12$  encoder layers, its hidden state  $h_{[\text{CLS}]} \in \mathbb{R}^{768}$  summarises the entire complaint and feeds the softmax classification head (Section 3.8). The [CLS] row accumulates context from every other token through the stacked encoder layers.

### 3.1.6 Pipeline summary

Figure 3.1 previews the end-to-end natural language processing (NLP) path from a free-text chief complaint to a SATS colour and pathway card. Each box corresponds to one transformation; the subsections and sections that follow derive the formulas, symbols, and worked steps for every stage.

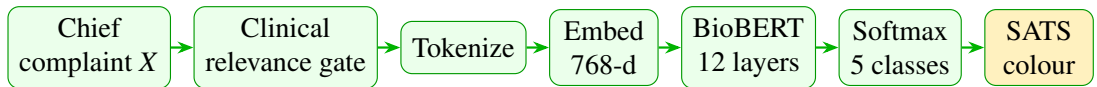


Figure 3.1: NLP pipeline from chief complaint to SATS colour (TEWS is documented separately in Section 3.3).

1. **Chief complaint  $X$ .** The patient or attendant submits one unstructured **English** text string describing symptoms and relevant history (Section 3.1.1). Tokenisation begins in Section 3.7.
2. **Clinical relevance gate.** A rule-based score  $g(X)$  filters greetings and non-medical text before BioBERT runs; see Section 3.17.
3. **Tokenize.** WordPiece splits  $X$  into subword tokens with special markers [CLS], [SEP], and <PAD>; see Section 3.7.

4. **Embed (768-d).** Each token ID maps to a 768-dimensional vector; positional and segment embeddings are added to form the input matrix  $H^{(0)}$ ; see Sections 3.9 and 3.10.
5. **BioBERT (12 layers).** Twelve encoder layers apply multi-head self-attention and feed-forward transforms; see Sections 3.11 and 3.12.
6. **Softmax (5 classes).** The [CLS] summary vector  $h_{[\text{CLS}]}$  passes through a classification head yielding acuity probabilities  $\hat{\mathbf{y}}$ ; see Section 3.13.
7. **SATS colour and pathway.** The predicted level  $\hat{c}$  maps to colour  $C_{\text{NLP}} = f_{\text{SATS}}(\hat{c})$  and pathway card  $P(C_{\text{NLP}})$ ; see Sections 3.15 and 3.16.

Each stage above is derived and exemplified in the referenced sections. Section 3.19 restates the full path for the implemented prototype.

### 3.1.7 Safety and nurse authority

The chatbot is designed as a **Flow Management Engine assist**, not as an autonomous discharge or treatment authority. Three design choices enforce that posture throughout Chapter 3:

1. **Gate before model.** Non-clinical text never receives a colour (Section 3.17).
2. **Softmax as a distribution.** Output is  $\hat{\mathbf{y}}$ , not a binary “safe/unsafe” certificate; low confidence and near ties escalate to nurses (Section 3.13).
3. **TEWS remains nurse-owned.** When vitals exist, TEWS and SATS discriminators govern bedside colour; the text path does not silently overwrite them (Section 3.3).

Pathway cards therefore propose destinations and timers that nurses can accept, modify, or reject. That principle is repeated in the results discussion (Section 4.6) and in the research questions on Softmax confidence (RQ7).

Table 3.1 records the mathematical object entering and leaving each stage so the reader can follow how one complaint string becomes a colour and pathway card.

| Stage           | Input               | Output                                                                     |
|-----------------|---------------------|----------------------------------------------------------------------------|
| Chief complaint | Patient text string | $X$ (raw string)                                                           |
| Clinical gate   | $X$                 | Pass/fail via $g(X)$                                                       |
| Tokenize        | $X$                 | Token sequence $(t_1, \dots, t_M)$ ,<br>$M \leq 128$                       |
| Embed           | Token IDs           | Input matrix $H^{(0)} \in \mathbb{R}^{M \times 768}$                       |
| BioBERT         | $H^{(0)}$           | Final matrix $H^{(12)}$ ; summary<br>$h_{[\text{CLS}]}$                    |
| Softmax         | $h_{[\text{CLS}]}$  | Acuity probabilities $\hat{\mathbf{y}} \in \mathbb{R}^5$                   |
| SATS colour     | $\hat{\mathbf{y}}$  | Level $\hat{c}$ , colour $C_{\text{NLP}}$ ,<br>pathway $P(C_{\text{NLP}})$ |

Table 3.1: Pipeline stage inputs and outputs (NLP path).

## 3.2 SATS colour chart

SATS assigns each living patient an urgency colour with a maximum waiting time  $T_{\max}$  and hospital destination (South African Triage Group, 2012). Figure 3.2 separates immediate and special protocols from the decreasing-urgency scale used for living patients. Blue indicates deceased-on-arrival dignity protocol, not low urgency.

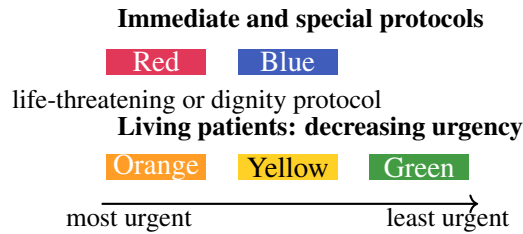


Figure 3.2: South African Triage Scale (SATS) colour bands used at hospital intake.

| Colour | Meaning               | $T_{\max}$ | Typical destination                               |
|--------|-----------------------|------------|---------------------------------------------------|
| Red    | Immediate life threat | 0 min      | Resuscitation bay                                 |
| Orange | Very urgent           | 10 min     | Acute / high-dependency bed                       |
| Yellow | Urgent                | 60 min     | Emergency waiting / urgent stream                 |
| Green  | Non-urgent            | < 4 h      | Outpatient department (OPD) / minors / polyclinic |
| Blue   | Dignity protocol      | N/A        | Mortuary / private space                          |

Table 3.2: SATS colours, waiting targets, and routing (SATS-aligned).

**Remark 3.1.** Red signals an immediate threat to life; Green indicates that the patient

*may safely wait in a routine stream. The chatbot outputs one of these colours so intake staff can route patients without re-interpreting free text.*

For the illustrative complaint in this chapter, the NLP model in Section 3.13 may assign moderate urgency (Yellow) with high probability; the same framework applies regardless of presenting symptoms.

### 3.3 TEWS: Triage Early Warning Score

TEWS is the deterministic score nurses compute from six vital parameters at SATS triage (South African Triage Group, 2012). It provides an objective colour band when measurements exist.

#### 3.3.1 Definition

Let  $\mathbf{v} = (v_1, \dots, v_6)$  denote HR, RR, mobility, temperature, AVPU consciousness, and trauma flag. Each parameter maps to points  $f_k(v_k) \in \{0, 1, 2, 3\}$  via SATS lookup tables. TEWS aggregates these contributions into one integer so that nurses at different desks can compare patients on a common scale. Weights  $w_k = 1$ :

$$T = \sum_{k=1}^6 w_k f_k(v_k) = \sum_{k=1}^6 f_k(v_k). \quad (3.1)$$

Summing six bounded terms yields a score  $T \in [0, 18]$  in principle, though typical values at triage are much lower. Because each  $f_k$  is defined from the SATS manual, the score is reproducible and auditable: two nurses who measure the same vitals obtain the same  $T$ .

| $k$ | Parameter $v_k$       | Source                  | Unit / scale                         |
|-----|-----------------------|-------------------------|--------------------------------------|
| 1   | Heart rate (HR)       | SATS manual, TEWS table | beats/min                            |
| 2   | Respiratory rate (RR) | SATS manual, TEWS table | breaths/min                          |
| 3   | Mobility              | SATS manual             | walking / assisted / immobile        |
| 4   | Temperature           | SATS manual             | °C                                   |
| 5   | AVPU consciousness    | SATS manual             | alert / verbal / pain / unresponsive |
| 6   | Trauma                | SATS manual             | present / absent                     |

Table 3.3: TEWS parameters and documentation source.

**Remark 3.2.** *Each vital sign contributes zero to three points. Elevated HR, abnormal RR, or reduced consciousness increase  $T$ ; normal values contribute zero.*

### 3.3.2 Piecewise scoring functions

The piecewise functions  $f_1$  and  $f_2$  encode tachycardia, bradycardia, and abnormal respiratory rate as higher point values, reflecting physiological stress seen at triage.

Heart rate  $f_1$  (beats/min):

$$f_1(v_1) = \begin{cases} 3 & v_1 \geq 130 \\ 2 & 111 \leq v_1 \leq 129 \text{ or } v_1 \leq 40 \\ 0 & 51 \leq v_1 \leq 100 \\ 1 & \text{otherwise} \end{cases} \quad (3.2)$$

Elevated heart rate (tachycardia) or very low heart rate therefore increases  $T$  and may raise the TEWS colour band.

Respiratory rate  $f_2$  (breaths/min):

$$f_2(v_2) = \begin{cases} 3 & v_2 \geq 30 \\ 2 & 21 \leq v_2 \leq 29 \\ 0 & 9 \leq v_2 \leq 14 \\ 1 & \text{otherwise} \end{cases} \quad (3.3)$$

Rapid breathing (tachypnoea) similarly adds points, capturing respiratory distress before full clinical assessment.

Mobility  $f_3$ : walking = 0; assisted = 1; immobile = 2–3 (SATS manual). Temperature  $f_4$ : 35.0–38.4°C = 0; borderline fever or hypothermia = 1; extreme values = 2–3. AVPU  $f_5$ : alert = 0, verbal = 1, pain = 2, unresponsive = 3. Trauma  $f_6$ : none = 0; major trauma = 2–3.

### 3.3.3 Colour map

The mapping  $C_{\text{TEWS}}(T)$  converts the integer score into the same colour vocabulary used at the SATS desk, so vital-sign triage aligns with hospital routing rules (South African Triage Group, 2012):

$$C_{\text{TEWS}}(T) = \begin{cases} \text{Red} & T > 7 \\ \text{Orange} & 5 \leq T \leq 6 \\ \text{Yellow} & 3 \leq T \leq 4 \\ \text{Green} & T \leq 2 \end{cases} \quad (3.4)$$

Thresholds in Equation (3.4) follow the standard SATS implementation: a single point

difference can change the colour band, so nurses record each  $f_k$  carefully. The diagram below visualises these bands on the score axis.

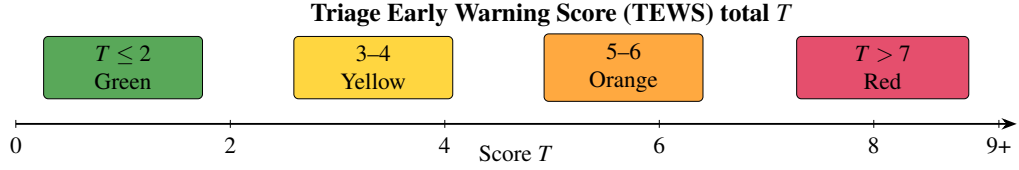


Figure 3.3: TEWS score ranges mapped to SATS colours (segments separated for readability).

### 3.3.4 Relationship to the text Softmax path

TEWS and BioBERT Softmax share a colour vocabulary (Red, Orange, Yellow, Green) but differ in inputs and mathematics:

- TEWS is a **weighted sum of six vital features**  $T = \sum_k w_k f_k(v_k)$  with  $w_k = 1$  in the standard adult score, then a piecewise colour map  $C_{\text{TEWS}}(T)$ .
- Softmax is a **Categorical distribution over five acuity levels** from complaint text, then the deterministic map  $f_{\text{SATS}}$ .

They are complementary, not competitors. At arrival, text may be the only signal; after nurse assessment, TEWS and clinical discriminators remain authoritative. This thesis therefore documents both, implements Softmax fully, and never treats TEWS absence as a chatbot limitation: nurses already compute TEWS when vitals exist.

| Aspect         | TEWS                       | BioBERT Softmax             |
|----------------|----------------------------|-----------------------------|
| Input          | Six vitals / flags $v_k$   | English chief complaint $X$ |
| Core map       | Integer sum $T$            | Probabilities $\hat{y}$     |
| Colour rule    | $C_{\text{TEWS}}(T)$ bands | $f_{\text{SATS}}(\hat{c})$  |
| When available | After measurement          | At text submission          |
| Authority      | Nurse desk standard        | Intake assist (overridable) |

Table 3.4: Side-by-side comparison of TEWS and the text Softmax path.

| $k$                          | Parameter   | Measured $v_k$ | $f_k(v_k)$ | Running sum |
|------------------------------|-------------|----------------|------------|-------------|
| 1                            | HR          | 125 bpm        | 2          | 2           |
| 2                            | RR          | 26 breaths/min | 2          | 4           |
| 3                            | Mobility    | walking        | 0          | 4           |
| 4                            | Temperature | 36.8 °C        | 0          | 4           |
| 5                            | AVPU        | alert          | 0          | 4           |
| 6                            | Trauma      | absent         | 0          | 4           |
| <b>Total <math>T</math>:</b> |             |                |            | <b>4</b>    |

Table 3.5: Term-by-term TEWS calculation for HR = 125 bpm, RR = 26 breaths/min, all other parameters normal.

**Example 3.1.** Suppose vitals are HR = 125 bpm and RR = 26 breaths/min; mobility, temperature, AVPU, and trauma are normal or absent. Table 3.5 lists each  $f_k$  contribution; summing gives

$$\begin{aligned}
f_1(125) &= 2 & (HR \text{ band } 111\text{--}129), \\
f_2(26) &= 2 & (RR \text{ band } 21\text{--}29), \\
f_k(\cdot) &= 0 & k = 3, \dots, 6, \\
T &= 2 + 2 = 4, \\
C_{\text{TEWS}}(4) &= \text{Yellow} \quad (\text{Equation (3.4)}).
\end{aligned}$$

If only these two vitals were measured, the score is partial but still auditable. At intake, before vitals exist, the BioBERT path in Section 3.13 supplies the text-based colour.

### 3.4 Neural networks, Transformers, and BERT

The text path of this project is implemented as a **neural network**: a composition of learnable maps that send numerical vectors to numerical vectors. Each map (a layer) has weights adjusted during training so that the overall composition predicts the correct acuity level from complaint text. Classical rule lists cannot capture the variety of English phrasing at intake; stacked nonlinear layers with attention can.

In the triage setting the input is already discrete after tokenisation (Section 3.7): integer IDs indexing an embedding table. The network never “reads English letters” directly; it reads vectors in  $\mathbb{R}^{768}$  and repeatedly transforms them. Training adjusts those transformations so that the final Softmax puts high probability on the nurse-assigned acuity level.

A **Transformer** (Vaswani et al., 2017) is a neural architecture whose central mechanism is *self-attention*: every token in a sequence can attend to every other token in one

parallel step. Unlike recurrent networks that read left-to-right only, a Transformer encoder builds context from the full window of tokens at once. That property matters for triage language, where negation and severity modifiers (“no pain”, “severe headache”) must reshape the meaning of neighbouring words. Scaled dot-product attention (Section 3.12) is the concrete formula used inside each BioBERT layer.

**BERT** (Bidirectional Encoder Representations from Transformers) (Devlin et al., 2019) is a deep Transformer *encoder* pre-trained with masked language modelling so that each token representation depends on left and right context. Masked language modelling hides random tokens and asks the model to recover them from surrounding biomedical or general text; the result is a reusable encoder rather than a chat generator. **BioBERT** (Lee et al., 2020) continues that pre-training on PubMed and PubMed Central text, then this project fine-tunes the encoder for five-class acuity prediction (Chapter 4).

Why BioBERT rather than a general BERT checkpoint alone? Domain continued pre-training places clinical vocabulary (“dyspnoea”, “malaria”, “epigastric”) in a representation space already shaped by biomedical corpora, so Stage 3 fine-tuning on chief complaints starts closer to triage language than a purely Wikipedia-trained encoder would.

The remainder of this chapter specifies the pipeline in order: English input  $X$ , clinical gate, WordPiece tokenisation  $\tau$ , three embedding components, twelve encoder layers, Softmax classification over mutually exclusive SATS levels, colour map  $f_{\text{SATS}}$ , and pathway card  $P(C)$ .

## 3.5 Introducing BERT

TEWS ignores language. A patient may describe symptoms in free text before HR or RR are recorded. Subjective complaints often appear first; bidirectional encoding is required so that “not feverish” and “very feverish” are not treated alike.

Vaswani et al. (2017) introduced scaled dot-product attention (Section 3.12). Devlin et al. (2019) stacked attention into deep encoders with MLM pre-training. The input representation preview is

$$E(t_i) = E_{\text{word}}(t_i) + E_{\text{pos}}(i) + E_{\text{seg}}(t_i), \quad (3.5)$$

where  $E_{\text{word}}$  identifies the token,  $E_{\text{pos}}$  encodes position  $i$ , and  $E_{\text{seg}}$  marks sentence segment (always segment 0 for a single complaint). Full detail appears in Section 3.9.



| Term                | Meaning                                    | Example in triage                  |
|---------------------|--------------------------------------------|------------------------------------|
| $E_{\text{word}}$   | Token identity: which word or subword?     | “headache” vs “feverish”           |
| $E_{\text{pos}}(i)$ | Position in the sequence: order matters    | “head ache” $\neq$ “ache head”     |
| $E_{\text{seg}}$    | Sentence segment (A or B in paired inputs) | Always segment 0 for one complaint |

Table 3.6: Breakdown of the three terms in Equation (3.5).

**Remark 3.3.** *BERT processes the full sentence simultaneously. The representation of “feel feverish” depends on “headache” and neighbouring tokens in both directions, which is essential for triage wording.*

## 3.6 Introducing BioBERT

BioBERT reuses BERT’s architecture (12 layers, 768 hidden dimensions, 12 attention heads) but continues MLM on biomedical corpora. Fine-tuning on nurse-labeled chief complaints adjusts the classification head so  $h_{[\text{CLS}]}$  maps to five acuity levels.

Clinical tokens such as “headache”, “feverish”, “aches”, and “weak” occupy regions of embedding space nearer related medical terms than in general BERT. After fine-tuning, their joint attention pattern supports acuity estimation for any presenting complaint, not only the illustrative example used in this chapter.

**Remark 3.4.** *Hospital intake is frequently text-first. BioBERT provides the mathematical bridge from unstructured complaint text to the same colour vocabulary nurses use with TEWS and SATS.*

## 3.7 Tokenisation

### 3.7.1 Input $X$

The NLP path begins with a single English *chief complaint* string  $X$ : the primary reason the patient presents, in free text. It is not a diagnosis, not a TEWS vital-sign vector, and not a full medical history. Non-clinical messages may be rejected by the relevance gate (Section 3.17) before tokenisation runs. Only clinical  $X$  proceeds into the encoder.

### 3.7.2 The tokenisation map $\tau$

Neural networks require fixed-size numerical input. Tokenisation is the first transformation after the gate: a map  $\tau$  sends raw complaint text  $X$  into a bounded

sequence of subword tokens with integer IDs.

$$X \xrightarrow{\tau} (t_1, \dots, t_M), \quad M \leq 128. \quad (3.6)$$

Here  $\tau$  is the **WordPiece tokenisation map** used by BERT and BioBERT: a *deterministic discrete encoding* that segments the string using a fixed vocabulary and emits token IDs (including special markers). It is **not** a neural network layer; it does not learn during fine-tuning. Learning begins at the embedding lookup that follows.

### 3.7.3 Why the maximum length is $M = 128$

The project fixes maximum sequence length at  $M = 128$  tokens because that is the BioBERT-base / deployment configuration used for training and inference. The cap has three roles:

- **Architecture match.** BioBERT was pre-trained and fine-tuned with this window; changing  $M$  would require a different checkpoint or expensive re-adaptation.
- **Batching.** Padding every complaint to length 128 yields rectangular tensors so many requests can share the same matrix multiply at intake.
- **Bound on context.** At most 128 subword positions (including [CLS], [SEP], and pads) enter one forward pass; longer free text must be truncated or summarised before submission.

**Remark 3.5.** *The fixed length  $M = 128$  is therefore both an architecture match and a practical batching choice: BioBERT was trained with this cap, and padding shorter complaints to 128 lets the hospital batch many requests through the same matrix operations at intake.*

**WordPiece** keeps common clinical words (“fever”, “pain”) whole and splits rare or complex forms (e.g. “headache”  $\rightarrow$  head+##ache). Vocabulary  $\mathcal{V}$  has  $|\mathcal{V}| \approx 28,996$  entries; each token ID indexes a row in the embedding table. When a word splits into two subwords, it occupies **two positions** in the sequence (see rows 4 and 7 in Table 3.8), so  $M$  counts subword tokens, not raw words.

Three special tokens structure every input sequence:

- [CLS] (ID 101) is inserted at position  $i = 0$  as the classification summary token; its final hidden state feeds the acuity head (Section 3.8).
- [SEP] (ID 102) is the **separator** token marking the end of the complaint sentence. It tells the model where clinical text stops, so subsequent <PAD> fillers are not interpreted as symptoms.

- $\langle \text{PAD} \rangle$  (ID 0) fills unused positions up to length  $M = 128$ ; the attention mask sets these positions to zero so they do not affect encoding.

| Token                        | ID  | Purpose                                 |
|------------------------------|-----|-----------------------------------------|
| [CLS]                        | 101 | Classification summary; always first    |
| [SEP]                        | 102 | End-of-sentence boundary before padding |
| $\langle \text{PAD} \rangle$ | 0   | Batch padding to fixed length 128       |

Table 3.7: Special tokens in the BioBERT tokenizer.

| $i$    | Raw      | Token                        | ID         |
|--------|----------|------------------------------|------------|
| 0      | n/a      | [CLS]                        | 101        |
| 1      | I        | i                            | 1045       |
| 2      | have     | have                         | 2031       |
| 3      | a        | a                            | 1037       |
| 4      | headache | head+##ache                  | 7994, 8772 |
| 5      | and      | and                          | 1998       |
| 6      | feel     | feel                         | 2514       |
| 7      | feverish | fever+##ish                  | 9643, 6804 |
| 8      | .        | .                            | 1012       |
| 9      | n/a      | [SEP]                        | 102        |
| 10–127 | n/a      | $\langle \text{PAD} \rangle$ | 0          |

Table 3.8: Token IDs for the illustrative complaint (padding to length 128 omitted).

### 3.8 The [CLS] and [SEP] tokens

Sequence classification requires one fixed-length vector summarising the entire complaint. Rather than averaging all token rows (which would weight filler words equally with symptoms), BERT introduces an artificial summary token [CLS] at position  $i = 0$ . The [SEP] token at the end of the real text marks the sentence boundary: encoding stops treating tokens after [SEP] as part of the clinical complaint, which matters once  $\langle \text{PAD} \rangle$  fillers occupy the remaining positions. Only the [CLS] row is passed to the classification head; [SEP] itself is not used for acuity prediction.

After  $L = 12$  encoder layers:

$$h_{[\text{CLS}]} = H_{0,:}^{(L)} \in \mathbb{R}^{768}. \quad (3.7)$$

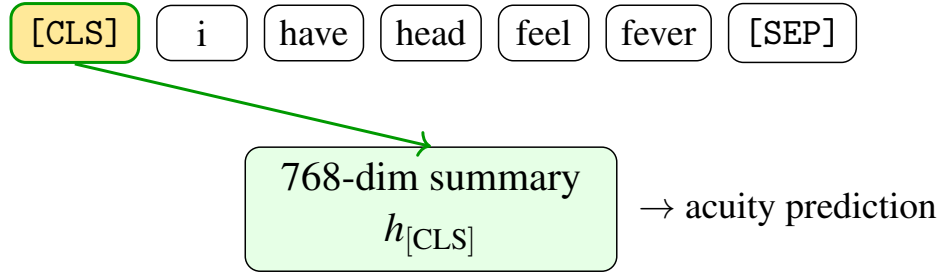


Figure 3.4: [CLS] aggregates context into one summary vector for classification.

Self-attention updates the [CLS] row at every layer so that it encodes information from all symptom tokens; the classification head therefore reads a single vector that represents the full clinical wording of  $X$ .

### 3.9 Embedding lookup

A token ID is an index, not a meaning. The embedding layer maps each ID to a dense vector in  $\mathbb{R}^{768}$  via the learned matrix  $E_{\text{word}} \in \mathbb{R}^{V \times 768}$ :

$$\mathbf{e}_i = E_{\text{word}}[\text{id}(t_i)] \in \mathbb{R}^{768}. \quad (3.8)$$

| Symbol         | Definition                                      |
|----------------|-------------------------------------------------|
| $V$            | Vocabulary size ( $\approx 30,000$ )            |
| $D = 768$      | Hidden dimension (BioBERT base configuration)   |
| $\mathbf{e}_i$ | Word embedding vector for token at position $i$ |

Table 3.9: Embedding lookup notation.

The lookup  $\mathbf{e}_i$  is one term in the full input representation  $E(t_i) = E_{\text{word}} + E_{\text{pos}} + E_{\text{seg}}$  from Equation (3.5). After PubMed pre-training, BioBERT places clinically related terms nearer in this space than general-purpose embeddings would.

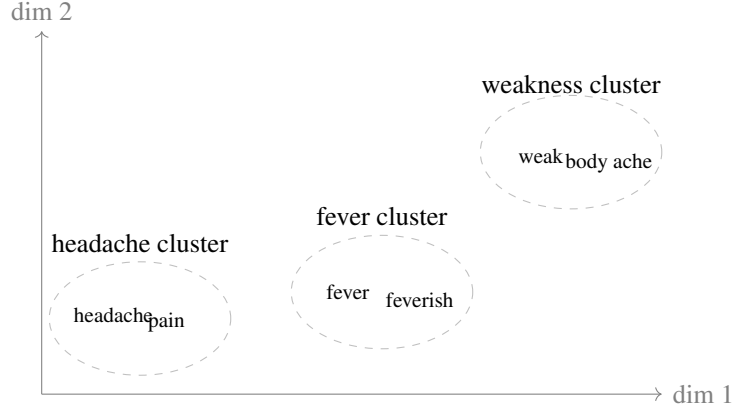


Figure 3.5: Schematic of BioBERT embedding space: familiar symptom words group together after biomedical pre-training (two examples per cluster).

When a patient writes “feverish” instead of “fever”, or “head ache” as two tokens, the model can still map the wording into regions of this space associated with similar presentations. That generalisation across varied patient language is central to text-first triage.

Contextual mapping sums three embeddings (Equation (3.5)). The positional embedding depends only on index  $i$ , and the segment embedding depends on segment id  $s$  (always  $s = 0$  for a single complaint):

$$E_{\text{pos}}(i) \in \mathbb{R}^{768}, \quad E_{\text{seg}}(s) \in \mathbb{R}^{768}, \quad s = 0. \quad (3.9)$$

The final input vector for token  $t_i$  at position  $i$  is the element-wise sum across all  $D = 768$  dimensions:

$$E(t_i)_d = E_{\text{word}}(t_i)_d + E_{\text{pos}}(i)_d + E_{\text{seg}}(s)_d, \quad d = 1, \dots, 768. \quad (3.10)$$

### 3.9.1 Why three embedding components

The model does not use a single lookup. Each token must pass through **three** additive components before the encoder stack, for distinct reasons:

1. **Word embedding**  $E_{\text{word}}$ . Encodes *what* the subword is (clinical meaning of “feverish” versus “football”).
2. **Position embedding**  $E_{\text{pos}}(i)$ . Encodes *where* the token sits in the complaint. Transformers have no built-in left-to-right recurrence; without position, “severe pain then mild” could not be distinguished from the reverse order.
3. **Segment embedding**  $E_{\text{seg}}(s)$ . Marks sentence A versus B in paired BERT tasks. For a single chief complaint we always use  $s = 0$ ; the term is kept so the BioBERT

checkpoint remains architecturally compatible even though a second segment is unused at intake.

Omitting word embeddings would remove meaning; omitting position would scramble order-sensitive clinical phrasing; omitting segment would break compatibility with the pre-trained BERT/BioBERT weight layout. The three components are therefore not decorative: they are the minimal input interface the Transformer expects.

### 3.9.2 Why the hidden dimension is $D = 768$

BioBERT-base uses hidden width  $D = 768$ . That width is fixed by the published architecture (Lee et al., 2020; Devlin et al., 2019) and matches

$$12 \text{ attention heads} \times 64 \text{ dimensions per head} = 768. \quad (3.11)$$

Every token vector, every encoder layer, and the classification summary  $h_{[\text{CLS}]}$  live in  $\mathbb{R}^{768}$ . The width is large enough to encode rich biomedical semantics yet small enough to stack twelve layers and train on large corpora. Changing  $D$  would require a different model family; this project therefore keeps  $D = 768$  throughout.

| Symbol                 | Definition                                      |
|------------------------|-------------------------------------------------|
| $E_{\text{word}}(t_i)$ | Row of $E_{\text{word}}$ looked up by token ID  |
| $E_{\text{pos}}(i)$    | Learned vector for position $i$ in the sequence |
| $E_{\text{seg}}(s)$    | Learned vector for segment $s$ (here $s = 0$ )  |
| $E(t_i)_d$             | Component $d$ of the summed embedding           |

Table 3.10: Symbols in the full input embedding (Equation (3.10)).

**Example 3.2.** For token “head” at position  $i = 4$ , suppose the first three dimensions (illustrative only) are

$$\begin{aligned}
E_{\text{word}} &= [0.31, -0.18, 0.52], \\
E_{\text{pos}}(4) &= [0.05, 0.02, -0.01], \\
E_{\text{seg}}(0) &= [0.00, 0.00, 0.00], \\
E(t_4) &= E_{\text{word}} + E_{\text{pos}}(4) + E_{\text{seg}}(0) \\
&= [0.36, -0.16, 0.51] \quad (\text{first three dimensions}).
\end{aligned}$$

The same element-wise addition repeats for all remaining dimensions  $d = 4, \dots, 768$ . The result is row 4 of  $H^{(0)}$ .

**Remark 3.6.** The embedding table is like a dictionary: each token ID points to a list of 768 numbers encoding meaning. Adding position tells the model where the word

appears; adding segment (zero here) reserves the mechanism for two-sentence inputs in other BERT tasks.

Figure 3.6 shows the flow for one token position. The 768 dimensions are a fixed width chosen in the original BERT architecture: wide enough to encode rich semantics, yet small enough to train twelve layers on large corpora.

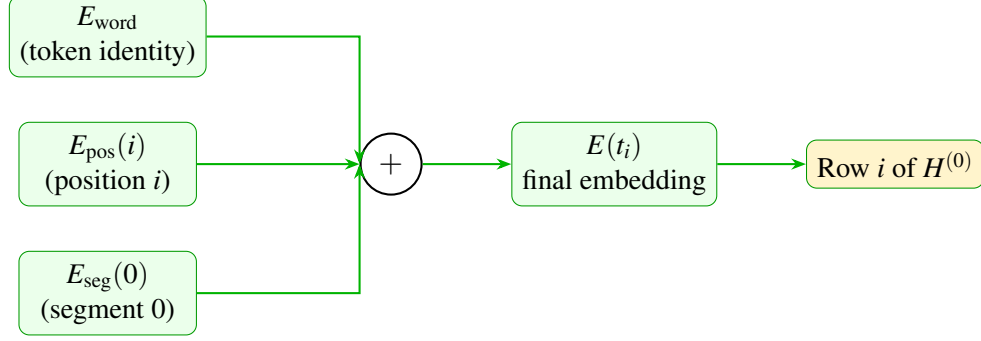


Figure 3.6: Three embeddings summed element-wise for one token position.

### 3.10 Input matrix $H^{(0)}$

Stacking the embedding vector for each of the  $M$  tokens forms the initial hidden-state matrix

$$H^{(0)} \in \mathbb{R}^{M \times 768}. \quad (3.12)$$

**Rows versus columns.** By convention in this thesis:

- **Rows** ( $M \leq 128$ ) index **token positions** in the sequence (row 0 is [CLS], then the WordPiece tokens, then [SEP] and pads).
- **Columns** (768) index **hidden feature dimensions** of the BioBERT representation.

Row  $i$  is built by applying Equation (3.10) at position  $i$ : look up the word vector, add the positional vector  $E_{\text{pos}}(i)$ , add the segment vector  $E_{\text{seg}}(0)$ , and stack the result. Viewed this way, a chief complaint is a table of numbers that the encoder layers progressively transform. The same row/column convention holds for  $H^{(\ell)}$  after each encoder layer: shape stays  $M \times 768$ ; only the entries change as self-attention mixes information *across rows*.

Later, the classification weight matrix  $W \in \mathbb{R}^{5 \times 768}$  uses the complementary layout: **rows** are the five acuity classes and **columns** are the 768 features that multiply  $h_{[\text{CLS}]}$  (Section 3.13).

| $i$      | Token    | $D_1$    | $D_2$    | $D_3$    | $D_4$    | $\dots$  |
|----------|----------|----------|----------|----------|----------|----------|
| 0        | [CLS]    | 0.02     | 0.11     | -0.05    | 0.33     | $\dots$  |
| 1        | i        | -0.08    | 0.04     | 0.19     | 0.01     | $\dots$  |
| 2        | have     | 0.15     | -0.22    | 0.07     | 0.44     | $\dots$  |
| 4        | head     | 0.31     | -0.18    | 0.52     | 0.09     | $\dots$  |
| 7        | fever    | 0.44     | 0.27     | -0.11    | 0.61     | $\dots$  |
| $\vdots$ | $\vdots$ | $\vdots$ | $\vdots$ | $\vdots$ | $\vdots$ | $\vdots$ |

Table 3.11: Example submatrix of  $H^{(0)}$  (first four of 768 dimensions; illustrative values).

After encoding, row 0 ([CLS]) holds the summary passed to the softmax classifier; the remaining rows are not used directly for the final colour decision.

### 3.11 Encoder layers

BioBERT applies the same encoder block  $L = 12$  times. Each block first lets tokens attend to one another (mixing symptom phrases), then applies a feed-forward network (FFN) to each row independently. **Residual connections** add the block input back to its output so that gradients can flow through depth during training. **Layer normalisation** rescales activations after each sub-step for numerical stability. The update equations are

$$Z^{(\ell)} = \text{LayerNorm}(H^{(\ell-1)} + \text{MultiHeadAttention}(H^{(\ell-1)})), \quad (3.13)$$

$$H^{(\ell)} = \text{LayerNorm}(Z^{(\ell)} + \text{FFN}(Z^{(\ell)})), \quad (3.14)$$

$$\text{FFN}(x) = W_2 \text{ReLU}(W_1 x + b_1) + b_2, \quad (3.15)$$

with  $W_1 \in \mathbb{R}^{768 \times 3072}$ ,  $W_2 \in \mathbb{R}^{3072 \times 768}$ , and bias vectors  $b_1 \in \mathbb{R}^{3072}$ ,  $b_2 \in \mathbb{R}^{768}$  in the standard BERT-base configuration. The inner dimension 3072 is four times the hidden width 768, giving the FFN enough capacity to transform each token row nonlinearly.

| Component          | What it does                                                                          |
|--------------------|---------------------------------------------------------------------------------------|
| MultiHeadAttention | Twelve parallel self-attention heads; each learns different word relationships        |
| FFN                | Two linear layers with ReLU: expands each row to 3072 dims, then projects back to 768 |
| LayerNorm          | Rescales activations for stable training across depth                                 |
| Residual (+)       | Adds sub-layer input to output so earlier information is not lost                     |

Table 3.12: Components inside one BioBERT encoder layer.



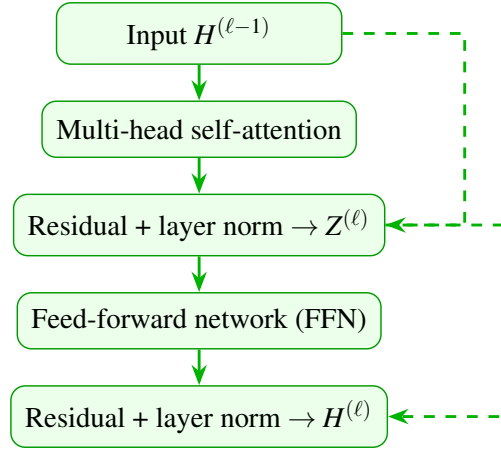


Figure 3.7: One BioBERT encoder layer (dashed arrows: residual connections). The block is stacked  $L = 12$  times.

In the attention sub-step, each token row is updated using information from every other row in the complaint, which is how “headache” and “feverish” influence one another before acuity is estimated. The FFN sub-step then applies the same nonlinear transformation to each row separately, increasing representational capacity. Repeating this block twelve times transforms  $H^{(0)}$  into  $H^{(12)}$ :

$$H^{(0)} \xrightarrow{\text{Layer 1}} H^{(1)} \xrightarrow{\text{Layer 2}} \dots \xrightarrow{\text{Layer 12}} H^{(12)} = H^{(L)}.$$

Shallow layers (1–4) tend to encode individual words and short phrases; middle layers (5–8) combine symptoms such as “headache” with “feverish” and “weak”; deep layers (9–12) encode broader urgency patterns. All  $M$  token rows are processed in parallel at each layer, not left-to-right as in older language models.

**Remark 3.7.** *Stacking layers is like reading a complaint once for words, once for symptom combinations, and once for overall urgency. The  $[CLS]$  row at layer 12 becomes  $h_{[CLS]}$  for classification.*

## 3.12 Self-attention

Self-attention computes a weighted combination of token representations for each position. The computation proceeds in four steps, matching Figure 3.8:

1. **Project** hidden rows into query, key, and value matrices.
2. **Score** pairwise compatibility between tokens.
3. **Normalise** scores into attention weights.
4. **Combine** value rows into context-aware outputs.

$$\begin{aligned}
Q &= HW_Q, \quad K = HW_K, \quad V = HW_V, \\
S &= QK^\top / \sqrt{d_k}, \\
A &= \text{softmax}(S), \\
\text{output} &= AV.
\end{aligned}$$

Following Vaswani et al. (2017), one attention head uses:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V, \quad (3.16)$$

with key dimension  $d_k = 64$  per head. For one query row, the softmax over keys assigns weight  $\alpha_j$  to token  $j$ :

$$\alpha_j = \frac{\exp(e_j)}{\sum_{k=1}^m \exp(e_k)}, \quad \sum_{j=1}^m \alpha_j = 1, \quad (3.17)$$

where  $e_j$  is the scaled score in row  $j$  of  $QK^\top / \sqrt{d_k}$  and  $m$  is the sequence length. BioBERT runs twelve such heads in parallel; their outputs are concatenated and projected back to 768 dimensions so each head can specialise (e.g. symptom co-occurrence vs negation).

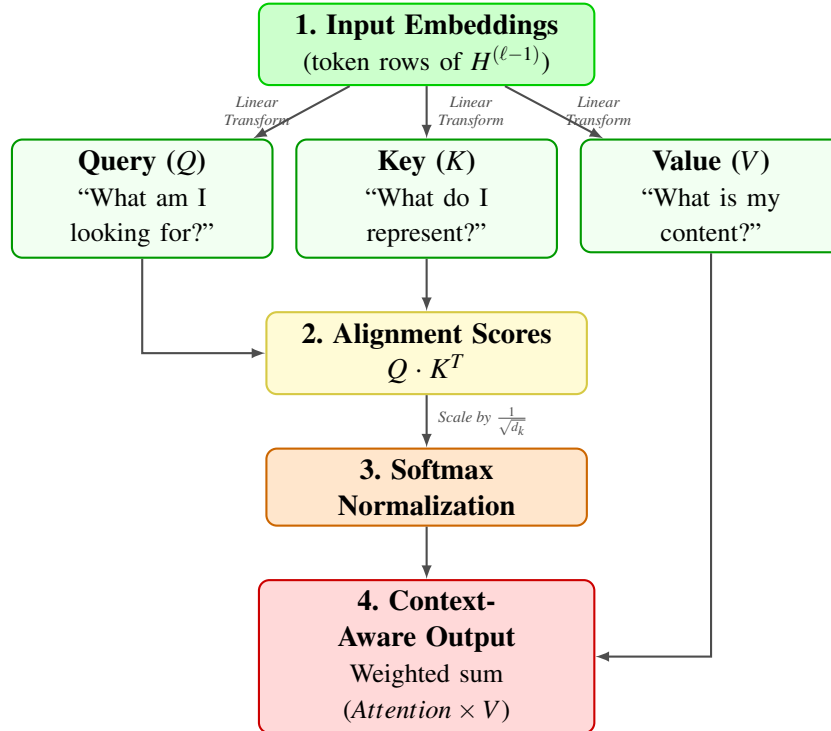


Figure 3.8: Self-attention data flow within one encoder layer.

| Symbol       | Definition                                                                        |
|--------------|-----------------------------------------------------------------------------------|
| $Q, K, V$    | Query, key, and value matrices derived from hidden states $H$                     |
| $QK^\top$    | Matrix of pairwise compatibility scores between tokens                            |
| $\sqrt{d_k}$ | Scaling constant ( $d_k = 64$ ) that prevents dot products from growing too large |
| $e_j$        | Raw attention score for token $j$ before normalisation                            |
| $\alpha_j$   | Normalised attention weight for token $j$ (a probability)                         |
| $m$          | Sequence length (number of tokens, at most 128)                                   |

Table 3.13: Self-attention notation.

| Query token  | [CLS] | headache | feverish | filler |
|--------------|-------|----------|----------|--------|
| Attention to | 0.09  | 0.32     | 0.41     | 0.18   |

Table 3.14: Schematic attention weights when the query is [CLS] (illustrative, not measured from the model). Symptom tokens receive higher weight than filler words.

The product  $QK^\top$  measures how strongly each token should attend to every other token. Dividing by  $\sqrt{d_k}$  keeps those scores in a range where the softmax in Equation (3.17) does not saturate. In a complaint mentioning both “headache” and “feverish”, attention typically assigns higher weight to symptom tokens than to function words such as “I” or “and”, so the updated representation emphasises clinically informative wording (Table 3.14).

## 3.13 Classification softmax

### 3.13.1 Mutually exclusive acuity categories

When a patient submits a chief complaint, their clinical urgency for routing is treated as falling into **exactly one** of the five SATS-aligned acuity levels  $\{1, 2, 3, 4, 5\}$ . A presentation cannot be Level 1 (immediate) and Level 5 (routine) at the same decision instant. Because only one level is selected for the pathway colour, the five categories are modelled as **mutually exclusive**. That modelling choice determines the output layer: the network must produce a distribution over five options, not five independent yes/no scores.

If the head used five independent sigmoid units, each class probability could be high at once (for example  $\hat{y}_1 = 0.9$  and  $\hat{y}_5 = 0.8$ ), which would not define a single routing colour. Softmax avoids that inconsistency by normalising across the five logits together.

### 3.13.2 Softmax as the probability translator

After BioBERT encoding, the classification head maps the summary vector  $h_{[\text{CLS}]} \in \mathbb{R}^{768}$  to five unconstrained real scores called **logits**:

$$\begin{aligned} \mathbf{z} &= W h_{[\text{CLS}]} + \mathbf{b}, \\ \hat{\mathbf{y}} &= \text{softmax}(\mathbf{z}), \\ \hat{y}_c &= \frac{\exp(z_c)}{\sum_{j=1}^5 \exp(z_j)}, \end{aligned} \tag{3.18}$$

with  $W \in \mathbb{R}^{5 \times 768}$  (**rows** = acuity classes, **columns** = features),  $\mathbf{b} \in \mathbb{R}^5$ , logits  $z_c$ , and  $\sum_{c=1}^5 \hat{y}_c = 1$ . Softmax enforces the two elementary rules of a finite probability distribution: every  $\hat{y}_c$  is non-negative, and the five values sum to one. The predicted level is  $\hat{c} = \arg \max_c \hat{y}_c$ . Expressing output as probabilities allows downstream routing logic to consider confidence (for example a dominant Yellow at about 69% versus a near tie between Yellow and Green).

This **classification Softmax** (over five acuity classes) is distinct from the **attention Softmax** inside each encoder layer (over token positions). Both use the same functional form; they act on different axes of the model. Attention Softmax asks “which tokens matter for this query?”; classification Softmax asks “which acuity level is most probable for this complaint?”

**Remark 3.8.** *The linear map  $\mathbf{z} = W h_{[\text{CLS}]} + \mathbf{b}$  is the triage analogue of a multi-class “score sheet”: each row of  $W$  emphasises features in  $h_{[\text{CLS}]}$  that favour one acuity level. Softmax then converts those raw scores into a comparable probability vector for hospital routing.*

### 3.13.3 Categorical distribution ( $K = 5$ )

By producing five probabilities that sum to one, Softmax parameterises a **Categorical distribution** over the acuity levels. A Categorical distribution is the discrete law for exactly  $K$  mutually exclusive outcomes; here  $K = 5$ . The vector  $\hat{\mathbf{y}}$  is the parameter of that law:

$$\hat{y}_c = P(\text{acuity} = c \mid X), \quad c \in \{1, 2, 3, 4, 5\}. \tag{3.19}$$

Informally,  $\hat{\mathbf{y}}$  behaves like a weighted five-sided die whose face probabilities are determined by the complaint text. Sampling from that die would draw a random acuity; hospital routing instead takes the mode  $\hat{c} = \arg \max_c \hat{y}_c$  (and may escalate to a nurse when the mode is not dominant).

Stage 3 training maximises the probability of the nurse-assigned level under this

Categorical model. Equivalently, it minimises the negative log-likelihood (NLL)

$$\mathcal{L}_{\text{NLL}} = -\log \hat{y}_y = -\log P(y | X), \quad (3.20)$$

which is exactly the cross-entropy loss used in Section 3.14 when labels are one-hot.

### 3.13.4 Bernoulli as the $K = 2$ building block

The **Bernoulli** distribution is the two-outcome special case ( $K = 2$ ): a single trial with results  $\{0, 1\}$  and success probability  $p$ . Softmax with two logits  $(z_0, z_1)$  yields

$$p = \frac{\exp(z_1)}{\exp(z_0) + \exp(z_1)} = \sigma(z_1 - z_0), \quad (3.21)$$

where  $\sigma$  is the logistic sigmoid. That identity shows why binary logistic regression is Softmax with  $K = 2$ . The five-class triage head is the same mathematical idea generalised from a weighted coin to five mutually exclusive faces.

**Example 3.3** (Bernoulli building block). *Suppose a toy two-class head distinguishes “urgent” versus “non-urgent” with logits  $z_U = 1.0$  and  $z_N = -0.5$ . Then*

$$\begin{aligned} \exp(z_U) &= e^{1.0} \approx 2.72, \\ \exp(z_N) &= e^{-0.5} \approx 0.61, \\ P(\text{urgent}) &= \frac{2.72}{2.72 + 0.61} \approx 0.82. \end{aligned}$$

*Raising  $z_U$  relative to  $z_N$  increases urgent probability exactly as raising one acuity logit relative to the others increases that class probability in the five-class Softmax.*

| Symbol             | Definition                                                       |
|--------------------|------------------------------------------------------------------|
| $h_{[\text{CLS}]}$ | 768-dimensional summary vector from final encoder layer          |
| $W, \mathbf{b}$    | Learned classification weight matrix ( $5 \times 768$ ) and bias |
| $z_c$              | Logit (pre-softmax score) for acuity level $c$                   |
| $\hat{y}_c$        | Categorical probability of level $c \in \{1, \dots, 5\}$         |
| $\hat{c}$          | Predicted level: index of largest $\hat{y}_c$                    |

Table 3.15: Classification head notation.

**Example 3.4** (Five-class Softmax for the running complaint). *Suppose illustrative logits*

for levels 1–5 are

$$\begin{aligned}\mathbf{z} &= [-1.2, -0.5, 1.8, 0.3, -0.9], \\ \exp(\mathbf{z}) &\propto [0.30, 0.61, 6.05, 1.35, 0.41], \\ \sum_{c=1}^5 \exp(z_c) &= 8.72, \\ \hat{\mathbf{y}} &\approx [0.03, 0.07, 0.69, 0.15, 0.05], \\ \hat{c} &= 3, \\ f_{\text{SATS}}(3) &= \text{Yellow}.\end{aligned}$$

Level 3 has the largest Categorical probability ( $\hat{y}_3 \approx 0.69$ , rounded to 72% in some deployment logs; Section 3.15). The mass on neighbouring Orange (0.07) and Green ( $0.15 + 0.05$ ) shows residual uncertainty: Softmax does not claim certainty, only a ranked distribution suitable for pathway assist with nurse oversight when confidence is low (RQ7 in Section 1.5).

**Remark 3.9.** Softmax probabilities are calibrated only to the extent that the training distribution matches live intake. High  $\hat{y}_c$  on Triagegeist holdout text is not a clinical guarantee at KNUST University Hospital; low-confidence or contradictory presentations remain mandatory nurse review.

## 3.14 Model training pipeline

The deployed triage classifier is not trained from random initialisation. It is the product of three sequential stages: biomedical pre-training (Stage 1), encoder transfer from a prior triage checkpoint (Stage 2), and project fine-tuning on nurse-labeled chief complaints (Stage 3). Stages 1 and 2 supply medical language and partial triage adaptation; Stage 3 is the comprehensive retraining that maps free-text complaints to five acuity levels for this chatbot.

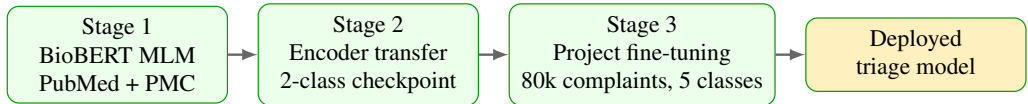


Figure 3.9: Three-stage training stack for the BioBERT triage classifier.

### 3.14.1 Stage 1: BioBERT pre-training (masked language modelling)

Before any triage labels are used, BioBERT is trained on biomedical text with masked language modelling (MLM). Random tokens are hidden and the model must predict

them from surrounding context in both directions:

$$\mathcal{L}_{\text{LM}} = - \sum_{i \in \mathcal{M}} \log P(t_i | H^{(L)}; \theta), \quad (3.22)$$

where  $\mathcal{M}$  is the set of masked positions,  $t_i$  is the true token at position  $i$ ,  $H^{(L)}$  is the final hidden-state matrix from all twelve layers,  $\theta$  denotes all model parameters, and  $P(t_i | \cdot)$  is the softmax probability assigned to the correct word. PubMed abstracts and PubMed Central full text ( $\approx 18$  billion words) teach medical vocabulary and syntax. Stage 1 does **not** teach triage urgency, SATS colours, or informal chief-complaint phrasing.

| Symbol                     | Definition                                        |
|----------------------------|---------------------------------------------------|
| $\mathcal{M}$              | Set of masked token positions in the sentence     |
| $t_i$                      | True (correct) word at masked position $i$        |
| $\theta$                   | All model weights and biases                      |
| $P(t_i   H^{(L)}; \theta)$ | Model probability for the correct token           |
| $\mathcal{L}_{\text{LM}}$  | Total MLM penalty; lower means better predictions |

Table 3.16: Masked language modelling notation (Stage 1).

**Example 3.5.** *If the masked token should be “headache”, the MLM contribution is*

*Good prediction ( $P = 0.92$ ):*

$$-\log(0.92) \approx 0.08 \quad (\text{small penalty}),$$

*Poor prediction ( $P = 0.01$ ):*

$$-\log(0.01) \approx 4.6 \quad (\text{large penalty}).$$

### 3.14.2 Stage 2: Encoder transfer from a triage checkpoint

A publicly available BioBERT triage checkpoint partially adapted the encoder to urgency-related text using a **two-class** classification head ( $W \in \mathbb{R}^{2 \times 768}$ ). For this project, that 2-class head is **discarded** on load because our labels require five acuity levels. The loader uses `ignore_mismatched_sizes=True`: only encoder weights carry forward; the output layer is randomly reinitialised for five classes. Stage 2 therefore contributes triage-oriented encoder features without constraining the final colour mapping.

### 3.14.3 Stage 3: Fine-tuning on nurse-labeled chief complaints

Stage 3 is the comprehensive retraining performed for this thesis. Nurse-assigned acuity labels from the Triagegeist benchmark (Section 3.1.2) are inner-joined with chief-complaint text on patient identifier, yielding  $N = 80,000$  English text-label pairs  $(X_i, y_i)$ .

Only `chief_complaint_raw` and `triage_acuity` are used; vitals and demographics in the source records are excluded so the NLP path matches the single-shot text input at intake.

The data are split 90/10 into training and validation sets ( $\approx 72,000$  /  $\approx 8,000$  rows, `test_size=0.1`, random seed 42). A new classification head  $W \in \mathbb{R}^{5 \times 768}$ ,  $\mathbf{b} \in \mathbb{R}^5$  is trained from scratch while encoder weights are updated with a small learning rate.

| Level            | Count         | Proportion   | SATS colour |
|------------------|---------------|--------------|-------------|
| 1 (most urgent)  | 3,222         | 0.0403       | Red         |
| 2                | 13,439        | 0.1680       | Orange      |
| 3                | 28,921        | 0.3615       | Yellow      |
| 4                | 23,020        | 0.2878       | Green       |
| 5 (least urgent) | 11,398        | 0.1425       | Green       |
| <b>Total</b>     | <b>80,000</b> | <b>1.000</b> |             |

Table 3.17: Class distribution in the 80,000 labeled chief-complaint training pairs.

| Hyperparameter              | Value                                  |
|-----------------------------|----------------------------------------|
| Epochs                      | 3                                      |
| Batch size (train and eval) | 16                                     |
| Learning rate $\alpha$      | $2 \times 10^{-5}$                     |
| Weight decay                | 0.01                                   |
| Max sequence length         | 128 tokens                             |
| Mixed precision (FP16)      | Enabled                                |
| Total optimizer steps       | 13,500 (3 epochs $\times$ 4,500 steps) |
| Best validation loss        | 0.001812 (epoch 3)                     |

Table 3.18: Fine-tuning hyperparameters (Stage 3).

| Component                           | Updated during Stage 3?           |
|-------------------------------------|-----------------------------------|
| Embedding layers                    | Yes (small learning-rate updates) |
| 12 transformer encoder layers       | Yes                               |
| Classification head $W, \mathbf{b}$ | Yes (trained from scratch)        |

Table 3.19: Which model components change during project fine-tuning.

The objective is **cross-entropy loss**: for each training example, the model is penalised when it assigns low probability to the nurse’s true acuity label. For a single example with true class  $y$ ,

$$\mathcal{L}_{\text{CE}} = -\log \hat{y}_y, \quad (3.23)$$



where  $\hat{y}_y$  is the probability the model assigns to the correct level. Over the full training set:

$$\mathcal{L}_{\text{triage}} = - \sum_{i=1}^N \log P(y_i | X_i; \theta), \quad N = 80,000. \quad (3.24)$$

**Example 3.6.** *If the nurse labeled a complaint Yellow (level 3), the cross-entropy penalty is*

*Low confidence on the true class ( $\hat{y}_3 = 0.05$ ):*

$$\mathcal{L}_{\text{CE}} = -\log(0.05) \approx 3.0 \quad (\text{large penalty}),$$

*High confidence on the true class ( $\hat{y}_3 = 0.95$ ):*

$$\mathcal{L}_{\text{CE}} = -\log(0.95) \approx 0.05 \quad (\text{small penalty}).$$

*Fine-tuning adjusts weights so correct classes receive higher probability on similar complaints.*

**Backpropagation** computes how each weight contributed to  $\mathcal{L}_{\text{triage}}$  by applying the chain rule backward from the loss through the softmax head and all twelve encoder layers. For classification weights  $W$ , with logits  $\mathbf{z} = Wh_{[\text{CLS}]} + \mathbf{b}$ ,

$$\frac{\partial \mathcal{L}}{\partial W} = \frac{\partial \mathcal{L}}{\partial \hat{\mathbf{y}}} \cdot \frac{\partial \hat{\mathbf{y}}}{\partial \mathbf{z}} \cdot \frac{\partial \mathbf{z}}{\partial W}. \quad (3.25)$$

In words: (i)  $\partial \mathcal{L} / \partial \hat{\mathbf{y}}$  measures how wrong the predicted probabilities are; (ii)  $\partial \hat{\mathbf{y}} / \partial \mathbf{z}$  propagates through softmax; (iii)  $\partial \mathbf{z} / \partial W$  links logits to the weight matrix. The same logic applies layer by layer through BioBERT.

**Gradient descent** then adjusts weights in the direction that reduces the loss:

$$W_{\text{new}} = W_{\text{old}} - \alpha \frac{\partial \mathcal{L}}{\partial W}, \quad (3.26)$$

with  $\alpha = 2 \times 10^{-5}$ . When the model confidently predicts the wrong colour, backpropagation increases the loss sharply and the subsequent weight update pushes similar complaints toward the nurse-labeled class on the next training pass.

**Remark 3.10.** *Pre-training (Stage 1) gives the model medical language; comprehensive fine-tuning (Stage 3) teaches triage decisions on short, informal chief-complaint phrasing. The learning rate is kept small so PubMed knowledge is preserved while the head and encoder adapt to urgency labels.*

### 3.15 Mapping acuity to colour: $f_{\text{SATS}}$

The softmax head predicts acuity levels 1–5 learned from training data. Hospital routing uses SATS colours, so a fixed deterministic map converts level to colour:

$$f_{\text{SATS}}(\hat{c}) = C_{\text{NLP}} = \begin{cases} \text{Red} & \hat{c} = 1 \\ \text{Orange} & \hat{c} = 2 \\ \text{Yellow} & \hat{c} = 3 \\ \text{Green} & \hat{c} \in \{4, 5\} \end{cases} \quad (3.27)$$

Levels 4 and 5 both correspond to non-urgent presentations in the training schema and therefore map to Green. Blue is reserved for dignity protocol on arrival and is not produced by this five-class head.

### 3.16 Clinical pathways $P(C_{\text{NLP}})$

Once  $C_{\text{NLP}}$  is known, the chatbot attaches a pathway card that specifies waiting time, destination, and escalation rules drawn from SATS-aligned hospital protocol at KNUST University Hospital:

$$P(C) = (T_{\text{max}}, \text{destination}, \text{actions}, \text{escalation}). \quad (3.28)$$

The four tuple components correspond directly to the columns in Table 3.20:  $T_{\text{max}}$  is the maximum waiting time, destination is the primary clinical department, actions list support steps (nursing, pharmacy, diagnostics), and escalation defines what happens if the timer expires or the patient worsens.

| Colour | $T_{\text{max}}$ | Destination                                            | Escalation                 |
|--------|------------------|--------------------------------------------------------|----------------------------|
| Red    | 0 min            | Resuscitation / Surgery                                | Immediate senior alert     |
| Orange | 10 min           | Internal Medicine / Surgery / Obs. & Gynae             | Head nurse if unclaimed    |
| Yellow | 60 min           | Outpatient urgent / Pediatrics                         | Re-assess if timer expires |
| Green  | < 4 h            | Outpatient / Dental / Eye / Public & Preventive Health | Return only if worse       |

Table 3.20: SATS pathway protocol at KNUST University Hospital (clinical destinations).

Table 3.21 is the authoritative full map: one row per NLP outcome (including gate rejection), with primary clinical departments, support actions, escalation rules, and an example complaint phrase.

| Scenario    | Trigger                | $T_{\max}$ | Primary clinical departments                        | Support actions                                                                                          | Escalation                   |
|-------------|------------------------|------------|-----------------------------------------------------|----------------------------------------------------------------------------------------------------------|------------------------------|
| Gate reject | $g(X) < 0.35$          | n/a        | Hold at intake                                      | Nursing: request complaint; HIT: no colour stored                                                        | Re-submit with clinical text |
| Red         | $\hat{c} = 1$          | 0 min      | Surgery, Internal Medicine                          | Nursing (vitals, TEWS), Pharmacy (emergency), Health Info (Red flag), Admin (bed bypass), HIT (Code Red) | Immediate senior clinician   |
| Orange      | $\hat{c} = 2$          | 10 min     | Internal Medicine, Surgery, Obs. & Gynae            | Nursing (monitor, timer), Diagnostics (stat labs), Pharmacy, Health Info, Admin, HIT                     | Head nurse if unclaimed      |
| Yellow      | $\hat{c} = 3$          | 60 min     | Outpatient urgent, Pediatrics, Diagnostics          | Nursing (vitals, TEWS, timer), Pharmacy, Health Info, HIT, Accounts (defer billing)                      | Re-assess at 60 min          |
| Green       | $\hat{c} \in \{4, 5\}$ | <4 h       | Outpatient, Dental, Eye, Public & Preventive Health | Nursing (brief triage), Diagnostics (routine), Health Info, HIT, Accounts                                | Return if worse              |

Table 3.21: Complete NLP pathway scenarios at KNUST University Hospital (gate reject plus four SATS colours).

| Scenario    | Example complaint phrase                                                     |
|-------------|------------------------------------------------------------------------------|
| Gate reject | “Hello, how are you?” (no clinical content)                                  |
| Red         | “Crushing chest pain, cannot breathe, sweating heavily”                      |
| Orange      | “Severe abdominal pain in pregnancy, bleeding”                               |
| Yellow      | <i>“I have a headache and feel feverish. My body aches and I feel weak.”</i> |
| Green       | “Routine dental pain for two weeks, no fever”                                |

Table 3.22: Example chief complaints triggering each pathway scenario.

Table 3.23 maps all hospital departments to their role in the pathway system. Clinical departments receive patients by colour; support departments execute actions attached to each pathway card.

| Department                      | Role     | Colours / function                          |
|---------------------------------|----------|---------------------------------------------|
| Internal Medicine               | Clinical | Red, Orange (acute medical)                 |
| Surgery                         | Clinical | Red, Orange (trauma, acute surgical)        |
| Obstetrics and Gynaecology      | Clinical | Orange (obstetric urgency)                  |
| Pediatrics                      | Clinical | Yellow (child urgent stream)                |
| Outpatient                      | Clinical | Yellow (urgent), Green (routine)            |
| Diagnostics Services            | Clinical | Yellow, Green (parallel labs/imaging)       |
| Dental                          | Clinical | Green                                       |
| Eye                             | Clinical | Green                                       |
| Public Health                   | Clinical | Green (community referrals)                 |
| Preventive Health               | Clinical | Green (screening, wellness)                 |
| Nursing                         | Support  | All colours (triage, vitals, reassessment)  |
| Pharmacy                        | Support  | All colours (medication on protocol)        |
| Health Information              | Support  | All colours (record colour and timer)       |
| Hospital Information Technology | Support  | All colours (digital routing, API)          |
| Administration                  | Support  | All colours (bed and queue management)      |
| Accounts                        | Support  | Green, Yellow (billing after stabilisation) |

Table 3.23: Department routing map at KNUST University Hospital.

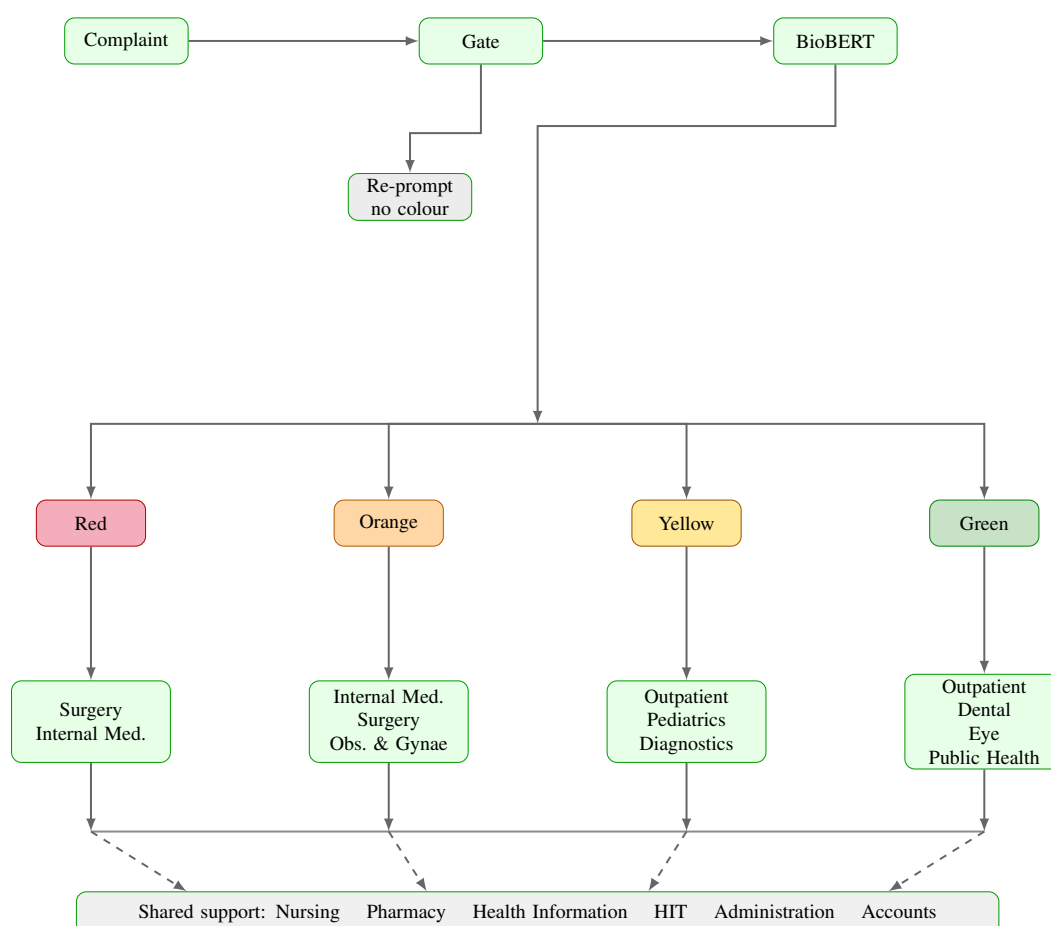


Figure 3.10: Complete NLP pathway map at KNUST University Hospital: gate rejection and all four SATS colours with clinical destinations and shared support services.

**Remark 3.11.** *Routing is protocol-driven; nurses may override the chatbot suggestion*

after clinical assessment. The chatbot supplies an initial colour and department direction before vitals are measured, reducing idle time at intake.

**Gate rejection.** When  $g(X) < 0.35$ , the pipeline stops before BioBERT (Table 3.21). Nursing staff request a proper chief complaint; no colour or pathway card is issued and Health Information stores no triage record until valid clinical text is submitted.

**Red** ( $\hat{c} = 1$ ). Immediate life-threatening presentations (e.g. crushing chest pain with dyspnoea) map to Red with  $T_{\max} = 0$ . Surgery and Internal Medicine activate resuscitation pathways; Nursing records immediate vitals and TEWS; Administration bypasses routine registration; Hospital Information Technology raises a Code Red alert.

**Orange** ( $\hat{c} = 2$ ). Very urgent cases such as severe abdominal pain in pregnancy route to Internal Medicine, Surgery, or Obstetrics and Gynaecology within ten minutes. Nursing starts a countdown timer; Diagnostics may run stat laboratory work; escalation to the head nurse occurs if no clinician claims the patient within  $T_{\max}$ .

**Yellow** ( $\hat{c} = 3$ ). The running illustrative complaint (“*I have a headache and feel feverish. My body aches and I feel weak.*”) maps here at approximately 72% confidence. Outpatient urgent stream, Pediatrics (when paediatric protocol applies), and parallel Diagnostics orders apply; Nursing sets a 60-minute reassessment timer and records the colour in Health Information.

**Green** ( $\hat{c} \in \{4, 5\}$ ). Non-urgent presentations (e.g. routine dental pain without fever) stream to Outpatient, Dental, Eye, or Public and Preventive Health within four hours. This diverts low-acuity patients away from acute waiting areas; Accounts handles billing after brief Nursing triage.

### 3.17 Medical relevance gate

Running BioBERT on non-clinical text (greetings, sports chat) would waste compute and produce meaningless acuity scores. A lightweight rule-based gate scores clinical relevance before the deep model runs:

$$g(X) = \text{clip}\left(\sum_j s_j(X) - p(X), 0, 1\right), \quad (3.29)$$

where  $s_j$  are non-negative feature scores from symptom lexicon hits and hospital-context phrases, and  $p(X) = 0.5$  when a non-clinical or greeting-only pattern matches.

Classification proceeds only if

$$g(X) \geq 0.35. \quad (3.30)$$

The threshold 0.35 was set in the prototype implementation to reject obvious non-medical input while accepting short but valid complaints (e.g. “chest pain”). Complaints that fail the gate return a request for clinical detail rather than a colour.

**Example 3.7.** *For the illustrative complaint with symptom hits “headache” ( $s_1 = 0.30$ ) and “feverish” ( $s_2 = 0.25$ ), with no greeting penalty ( $p(X) = 0$ ):*

$$\begin{aligned} \sum_j s_j &= 0.30 + 0.25 = 0.55, \\ g(X) &= \text{clip}(0.55 - 0, 0, 1) = 0.55 \geq 0.35 \end{aligned}$$

*so classification proceeds. For “Hello, how are you?” with no symptom hits and greeting penalty  $p(X) = 0.5$ :*

$$\begin{aligned} \sum_j s_j &= 0, \\ g(X) &= \text{clip}(0 - 0.5, 0, 1) = 0 < 0.35 \end{aligned}$$

*and the system requests a chief complaint instead of running BioBERT.*

### 3.18 Worked numerical walkthrough

This section traces one complaint through every stage of the NLP pipeline in Figure 3.1 and Section 3.19 with concrete numbers, so the Softmax and pathway sections are not left abstract. Let

$$X = \text{“I have a headache and feel feverish. My body aches and I feel weak.”}$$

as elsewhere in the thesis.

**Step 1: Gate.** Symptom lexicon hits for “headache” (0.30) and “feverish” (0.25), with no greeting penalty, give  $g(X) = 0.55 \geq 0.35$  (Section 3.17). Classification proceeds.

**Step 2: Tokenisation.** WordPiece yields a short sequence with [CLS], symptom subwords, [SEP], and pads to length  $M = 128$ . Only the first few positions carry clinical content; pads are masked in attention. The map  $\tau$  is deterministic: the same string always produces the same IDs.

**Step 3: Embeddings and  $H^{(0)}$ .** Each of the  $M$  positions receives a 768-dimensional vector

$$H_{i,:}^{(0)} = E_{\text{word}}(t_i) + E_{\text{pos}}(i) + E_{\text{seg}}(0).$$

Shape  $H^{(0)} \in \mathbb{R}^{128 \times 768}$ : rows are tokens, columns are features (Section 3.10).

**Step 4: Twelve encoder layers.** Self-attention mixes information across rows. After layer  $\ell = 12$ , row 0 is the summary  $h_{[\text{CLS}]} \in \mathbb{R}^{768}$ . Schematic attention (Table 3.14) illustrates higher weight on symptom tokens than on fillers.

**Step 5: Softmax.** Using the illustrative logits of Section 3.13,

$$\hat{\mathbf{y}} \approx [0.03, 0.07, 0.69, 0.15, 0.05], \quad \hat{c} = 3.$$

This is a Categorical distribution with mode Level 3.

**Step 6: Colour and pathway.**

$$C_{\text{NLP}} = f_{\text{SATS}}(3) = \text{Yellow}.$$

The pathway card is

$$P(\text{Yellow}) = (60 \text{ min, Outpatient urgent / Pediatrics, actions, escalation})$$

as in Table 3.21. Nursing sets a 60-minute reassessment timer; Diagnostics may run in parallel.

**Step 7: Parallel TEWS (when vitals arrive).** Suppose later measurements give TEWS integer  $T = 4$ . Then  $C_{\text{TEWS}}$  is Yellow on the SATS vital bands (Section 3.3). Text colour and vital colour agree in this illustration; if they disagreed, nurse judgment and SATS discriminators govern, not automatic Softmax override.

The walkthrough shows why every mathematical object in Chapter 3 appears in the deployed API response: gate metadata,  $\hat{\mathbf{y}}$ ,  $C_{\text{NLP}}$ , and pathway fields.

### 3.19 End-to-end summary

For any chief complaint  $X$ , the implemented text path (matching the prototype /predict endpoint) proceeds as follows:

$$\begin{aligned}
 X &\xrightarrow[3.17]{g} \tau(X) \xrightarrow[3.7-3.10]{} H^{(0)} \\
 &\xrightarrow[3.11-3.12]{12} h_{[\text{CLS}]} \xrightarrow[3.13]{} \hat{\mathbf{y}} \xrightarrow[3.15]{} C_{\text{NLP}} \xrightarrow[3.16]{} P(C_{\text{NLP}}),
 \end{aligned} \tag{3.31}$$

where each arrow label points to the section that defines that transformation. If  $g(X) < 0.35$ , the chain stops after the gate and no colour is assigned.

1. **Clinical relevance gate.** Compute  $g(X)$ ; if  $g(X) < 0.35$ , return a request for clinical detail and stop.
2. **Tokenisation and embedding.** Map  $X$  to tokens, build  $H^{(0)} \in \mathbb{R}^{M \times 768}$  (Sections 3.7–3.10).
3. **BioBERT encoding.** Apply  $L = 12$  encoder layers with self-attention (Sections 3.11–3.12) to obtain  $h_{[\text{CLS}]}$ .
4. **Classification.** Apply the softmax head (Section 3.13):

$$\begin{aligned}
 \hat{\mathbf{y}} &= \text{softmax}(Wh_{[\text{CLS}]} + \mathbf{b}), \\
 \hat{c} &= \arg \max_c \hat{y}_c.
 \end{aligned}$$

5. **Colour and pathway.** Apply  $C_{\text{NLP}} = f_{\text{SATS}}(\hat{c})$  and return pathway card  $P(C_{\text{NLP}})$  (Sections 3.15–3.16).

The illustrative complaint in this chapter instantiates the same pipeline; one worked outcome is level 3 (Yellow) at 72% confidence for headache and feverish symptoms. TEWS (Section 3.3) remains the nurse-facing vital-sign score when measurements become available.

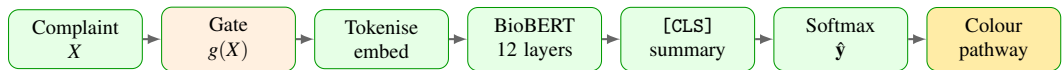


Figure 3.11: Compact end-to-end NLP triage path (TEWS documented separately in Section 3.3).

The diagram above matches the pipeline in Section 3.1. In order:

1. **Complaint**  $X$  enters as free text from the patient or attendant.
2. **Gate** computes  $g(X)$ ; non-clinical input is rejected before BioBERT runs.



3. **Tokenise and embed** map text to integer IDs and build the matrix  $H^{(0)}$ .
4. **BioBERT** applies twelve encoder layers with self-attention across all tokens.
5. **[CLS] summary** extracts the fixed-length vector  $h_{[\text{CLS}]}$  from row 0.
6. **Softmax** outputs acuity probabilities  $\hat{\mathbf{y}}$  over five classes.
7. **Colour and pathway** apply  $f_{\text{SATS}}(\hat{c})$  and return the pathway card  $P(C_{\text{NLP}})$ .

## 3.20 Implementation note

Figure 3.12 shows the deployable system: patient-facing web interface, machine-learning API, and hospital routing layer. The NLP mathematics in Figures 3.1 and 3.11 execute inside the API; department routing applies KNUST University Hospital pathway rules after classification.

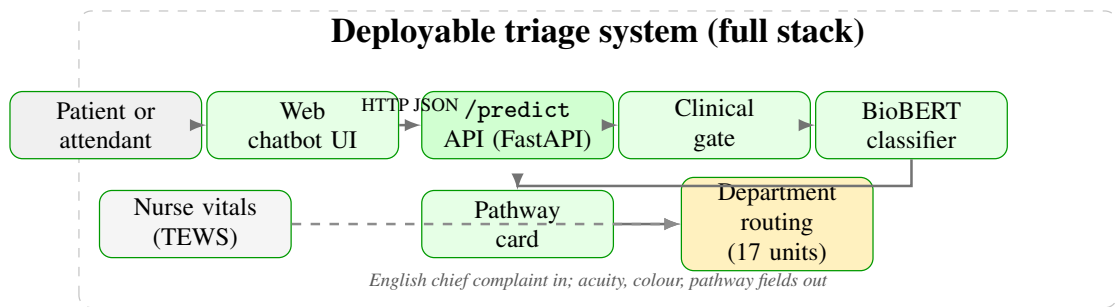


Figure 3.12: System architecture: user interface, `/predict` API, BioBERT inference pipeline, pathway card, and department routing at KNUST University Hospital. Nurse TEWS runs in parallel (dashed); prototype code in Appendix 5.4.

The prototype exposes a `/predict` application programming interface (API) endpoint: JSON **English** chief complaint in; acuity probabilities,  $C_{\text{NLP}}$ , pathway fields, and gate metadata out. Model weights and tokenizer files ship with the repository listed in Appendix 5.4. This thesis treats the mathematics above as the specification; empirical results appear in Chapter 4.

## CHAPTER 4

### RESULTS

This chapter begins with the **model itself**: BioBERT architecture constants, the three-stage training stack, and evidence that Stage 3 fine-tuning is required before the encoder can support safe acuity prediction. Only after that model analysis does the chapter report holdout classification metrics (baseline and SMOTE-augmented variants), inference latency, pathway routing at KNUST University Hospital, and anticipated impact on hospital workflow. Holdout numbers come from the stratified 10% holdout ( $N_{\text{eval}} = 8,000$ ). Expected operational benefits are interpreted from classification outcomes, latency, and published triage literature (KATH Emergency Department, 2018; Mitchell et al., 2023); prospective time-motion data at KNUST University Hospital are not yet available.

#### 4.1 BioBERT model analysis

The first results question is not “how accurate is the chatbot on a table?” but “what model is deployed, and does it work only after the right training?” Before project-specific fine-tuning, the classifier rests on BioBERT-base (Lee et al., 2020): a twelve-layer bidirectional encoder with hidden dimension 768, twelve attention heads, maximum sequence length 128 tokens, and a WordPiece vocabulary of approximately 29,996 tokens (Section 3.11). Stages 1 and 2 (Section 3.14) supply biomedical language understanding and partial triage-oriented encoder features; they do *not* produce the five-class acuity mapping required at intake. Stage 3 reinitialises the classification head and trains on the Triagegeist English corpus (Section 3.1.2). Class imbalance in the training corpus (Level 3 is the mode; Level 1 is the minority) motivates Stage 3 mitigation strategies documented in Section 4.2; it is not the focus of this section.

Figure 4.1 summarises the three-stage training stack. Stage 1 applies masked language modelling on PubMed and PubMed Central text; Stage 2 transfers encoder weights from a prior two-class triage checkpoint with the old head discarded; Stage 3 is the comprehensive project fine-tuning that maps free-text complaints to five SATS acuity levels.

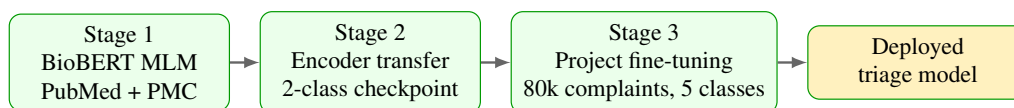


Figure 4.1: Three-stage BioBERT training stack (Stages 1–2 pre-date project fine-tuning; Stage 3 produces the deployed five-class head).

### Interpretation.

- **Chart type:** horizontal flow diagram (left-to-right pipeline).
- **Layout:** each box is one training stage; arrows show parameter flow.
- **Data source:** project training configuration (Chapter 3, Section 3.14).
- **Key readings:** Stage 1 optimises masked language modelling only; Stage 2 transfers encoder weights without the obsolete two-class head; Stage 3 is the only stage that maps chief complaints to five acuity levels.
- **Takeaway:** the chatbot’s urgency signal depends on completing Stage 3; Stages 1–2 alone cannot route patients safely.

Figure 4.2 contrasts expected accuracy *before* Stage 3 fine-tuning with the holdout result after fine-tuning. Each bar represents a different predictor; definitions are as follows:

- **Uniform random.** Selects one of five acuity levels with equal probability (20% expected accuracy). No learning; a sanity-check lower bound.
- **Random head (encoder only).** BioBERT encoder weights are loaded from Stages 1–2 but the five-class linear head is *untrained*. Tests whether biomedical pre-training alone can classify triage (expected  $\approx 20\%$ , same as chance).
- **Majority-class bound.** Always predicts Level 3 (Yellow), the training-set mode. Achieves 36.15% accuracy on this corpus but 0% Red recall; a clinically unsafe ceiling for naive rules.
- **Fine-tuned BioBERT.** Full Stage 3 checkpoint evaluated on the held-out 8,000 rows excluded from training (same split as Table 4.1). This is the only *measured* bar; the first three are theoretical or bound values.

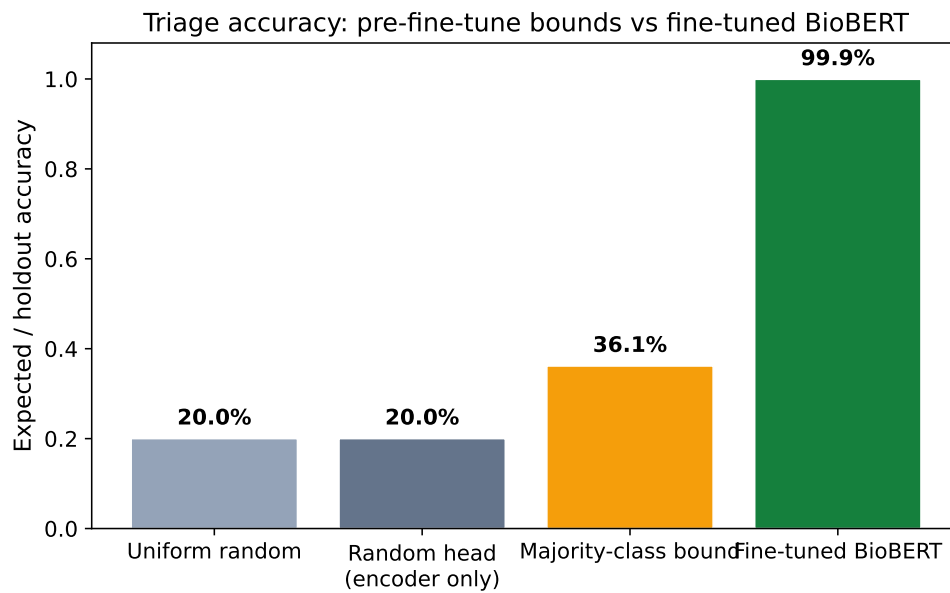


Figure 4.2: Expected or holdout accuracy: pre-fine-tune bounds vs fine-tuned BioBERT on the 8,000-row holdout set.

### Interpretation.

- **Chart type:** grouped bar chart.
- **Axes:** vertical axis = accuracy (0–1); horizontal axis = predictor type (four bars).
- **Data source:** theoretical bounds for the first three bars; measured holdout accuracy for fine-tuned BioBERT ( $N = 8,000$ ).
- **Key readings:** uniform random 20.0%, random head  $\approx 20\%$ , majority-class bound 36.15%, fine-tuned BioBERT 99.92%.
- **Takeaway:** encoder pre-training alone is insufficient for safe triage routing; Stage 3 fine-tuning on nurse-labeled complaints is required.

Figure 4.3 records the architecture constants used in this project.

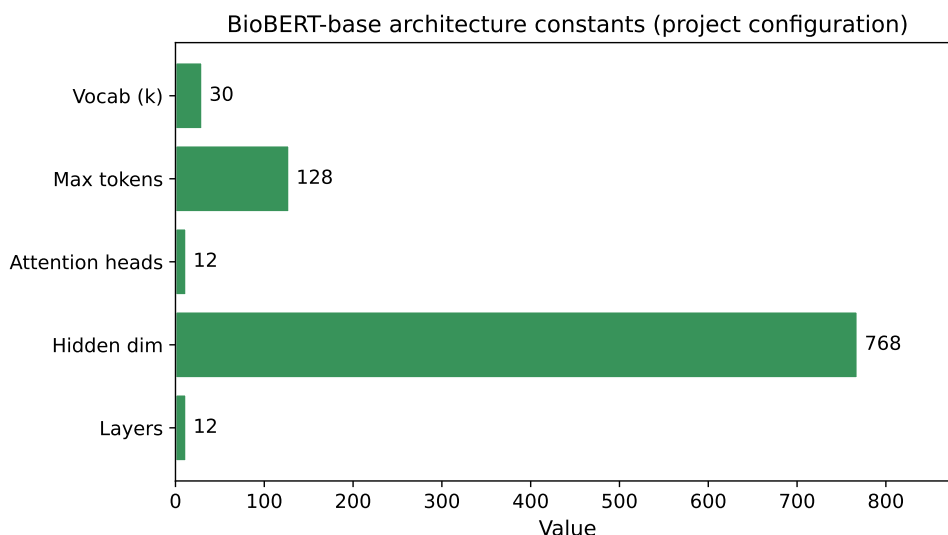


Figure 4.3: BioBERT-base architecture constants (project configuration).

### Interpretation.

- **Chart type:** horizontal bar chart.
- **Axes:** vertical axis = architecture parameter name; horizontal axis = numeric value (unitless counts or dimensions).
- **Data source:** project config.json and BioBERT-base specification (Lee et al., 2020).
- **Key readings:** 12 layers, 768 hidden units, 12 attention heads, 128 max tokens,  $\approx 30k$  vocabulary.
- **Takeaway:** the 128-token limit bounds how much complaint history can enter a single inference call at intake.

## 4.2 Fine-tuned model analysis

Having established the BioBERT configuration, the three-stage training stack, and the necessity of Stage 3 fine-tuning for triage routing, the remainder of this chapter reports quantitative holdout performance and operational implications.

### 4.2.1 Evaluation protocol

The deployed classifier is the product of Stage 3 fine-tuning documented in Section 3.14.3. Two checkpoints are compared on the *same* stratified 90/10 holdout split ( $N_{\text{eval}} = 8,000$ ):

1. **Baseline fine-tuned BioBERT** (project default): uniform sampling, three epochs, learning rate  $2 \times 10^{-5}$ , best validation cross-entropy 0.001812.
2. **SMOTE-augmented variant**: stratified splitting plus synthetic oversampling of minority urgent classes to improve Level 1 recall at possible cost to overall accuracy.

Metrics reported: accuracy, macro-F1, per-class precision/recall/F1, and inference latency (p50, p95, mean milliseconds per sample). Evaluation uses the same complaint texts and nurse labels as training; results measure *historical text classification*, not live patient outcomes.

| Metric                   | Baseline | SMOTE  |
|--------------------------|----------|--------|
| Accuracy                 | 0.9992   | 0.9985 |
| Macro-F1                 | 0.9977   | 0.9954 |
| Recall Level 1 (Red)     | 0.9814   | 1.0000 |
| Precision Level 1 (Red)  | 1.0000   | 0.9641 |
| Latency p50 (ms/sample)  | 100.30   | 95.50  |
| Latency p95 (ms/sample)  | 139.11   | 125.12 |
| Latency mean (ms/sample) | 102.13   | 90.17  |

Table 4.1: Holdout classification and latency metrics (stratified 10% split,  $N = 8,000$ ).

#### 4.2.2 Interpreting high holdout accuracy

Holdout accuracy above 99% can appear implausibly high without context. The following points clarify what the numbers represent and what they do *not* prove.

**Task definition.** The model predicts nurse-assigned acuity from chief-complaint text on a retrospective, in-distribution holdout ( $N = 8,000$ ) drawn from the Triagegeist synthetic English benchmark used for Stage 3 fine-tuning (Section 3.1.2). This is not external validation on a different hospital, time period, or patient population; KNUST-informed examples used for pathway illustration are excluded from these metrics.

**Label source.** Labels come from structured triage records (`chief_complaint_raw` inner-joined with `triage_acuity`). Because complaints and colours were recorded in the same clinical workflow, complaint text often encodes acuity explicitly (e.g. “crushing chest pain” vs “routine dental check”). High agreement with nurse labels therefore measures *label consistency on historical text*, not independent clinical correctness (Stewart et al., 2023).

**What is not measured.** These metrics do not establish prospective patient safety, waiting-time reduction, or physician-confirmed diagnosis. They quantify offline

agreement between model output and nurse-assigned colour on held-out rows from the training distribution.

**Residual errors.** The baseline model is not perfect: it misses 6 of 322 Red cases (98.14% Level 1 recall). Under-triage of urgent presentations remains the primary deployment risk (Berkowitz et al., 2019). The SMOTE variant achieves 100% Red recall but lowers Red precision to 96.41%, introducing false urgent alerts.

**Literature context.** NLP triage systems evaluated on structured complaint text in similar retrospective settings can report high agreement with nurse labels (Stewart et al., 2023; Gligorijević et al., 2018). Such results indicate that text carries acuity signal before vitals are recorded; they do not replace prospective audit.

**Generalisation caveat.** The prototype accepts **English** chief complaints only; holdout metrics reflect English text from the Triagegeist training corpus. Atypical phrasing or presentations absent from that benchmark may reduce agreement at live intake at KNUST University Hospital. Deployment should treat low-confidence or contradictory presentations as mandatory nurse review (South African Triage Group, 2012).

**Operational reading of Softmax.** Because  $\hat{y}$  is a Categorical distribution, the nurse-facing display should show more than the mode colour. A useful rule of thumb for assistive deployment is:

- **Dominant mode** ( $\hat{y}_c \geq 0.70$ ): route with the pathway card; nurse may still override.
- **Moderate confidence** ( $0.50 \leq \hat{y}_c < 0.70$ ): show colour and the next-highest competing level; flag for early nurse review.
- **Near tie** ( $\hat{y}_c < 0.50$  or two levels nearly equal): do not treat the chatbot colour as decisive; escalate to nurse assessment immediately.

These thresholds are operational heuristics for RQ7, not trained decision rules; prospective calibration at KNUST University Hospital may revise them.

### 4.2.3 SMOTE for class imbalance

Synthetic Minority Over-sampling Technique (SMOTE) generates synthetic training examples for under-represented classes by interpolating between existing minority samples in feature space (Chawla et al., 2002). Given two minority feature vectors  $\mathbf{x}_a$  and  $\mathbf{x}_b$ , a synthetic point is

$$\mathbf{x}_{\text{new}} = \mathbf{x}_a + \lambda(\mathbf{x}_b - \mathbf{x}_a), \quad \lambda \sim \text{Uniform}(0, 1). \quad (4.1)$$

Raw chief-complaint text cannot be SMOTEd directly because strings are discrete and interpolation between token sequences is undefined. This project therefore applies

imbalance mitigation in a text-classification pipeline:

1. **Stratified split.** Training and holdout partitions preserve class proportions (seed 42).
2. **Minority oversampling.** Urgent levels are duplicated or augmented so gradient updates are not dominated by Level 3.
3. **Text augmentation.** Paraphrase-style perturbations increase minority-class diversity without changing labels.
4. **SMOTE-augmented checkpoint.** Synthetic oversampling in the embedding or feature space of the training pipeline produces the SMOTE variant evaluated below.

The trade-off is visible on Level 1 (Red): SMOTE raises recall from 98.14% to 100.00% on holdout but lowers precision from 1.0000 to 0.9641 (a small number of non-Red complaints are upgraded to Red). Overall accuracy decreases slightly (99.92% to 99.85%). The baseline checkpoint remains the project default unless clinical stakeholders prioritise zero missed Red cases (Berkowitz et al., 2019).

#### 4.2.4 Baseline fine-tuned model

On the holdout set, baseline BioBERT achieves 99.92% accuracy and 0.9977 macro-F1 (Table 4.1). Level 1 (Red) recall is 98.14% with perfect precision (1.0000): no holdout case labeled Red was predicted as a lower-urgency level, and no false Red alarms were raised. Section 4.2.2 explains why these values reflect label agreement on in-distribution data rather than infallible clinical judgment.

Figure 4.4 shows the confusion matrix. Off-diagonal mass is negligible; the six missed Level 1 cases (322 support, 98.14% recall) represent the primary residual safety concern for deployment review.



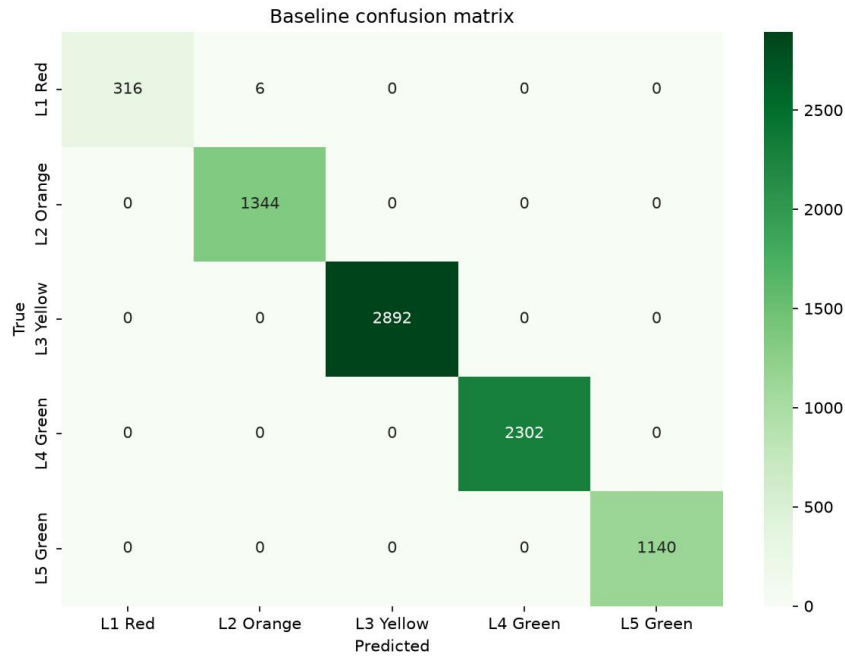


Figure 4.4: Baseline fine-tuned BioBERT confusion matrix (holdout,  $N = 8,000$ ). Rows = true nurse label; columns = model prediction.

#### Interpretation.

- **Chart type:** heatmap confusion matrix.
- **Axes:** rows = true nurse-assigned acuity (L1–L5); columns = model-predicted acuity; cell colour intensity = number of holdout cases in each true–predicted pair.
- **Data source:** baseline checkpoint, stratified holdout  $N = 8,000$ .
- **Key readings:** diagonal dominance (correct predictions); six off-diagonal cells where true Red was predicted as lower urgency.
- **Takeaway:** chief-complaint text carries strong acuity signal after Stage 3 fine-tuning, but the six missed Red cases require nurse override in deployment (Berkowitz et al., 2019).

#### 4.2.5 SMOTE-augmented variant

SMOTE augmentation raises Level 1 recall to 100% on holdout (all 322 Red cases correctly identified) at the cost of a small accuracy decrease (99.85%) and lower Red precision (96.41%): a small number of non-Red complaints are upgraded to Red. Macro-F1 remains above 0.995.

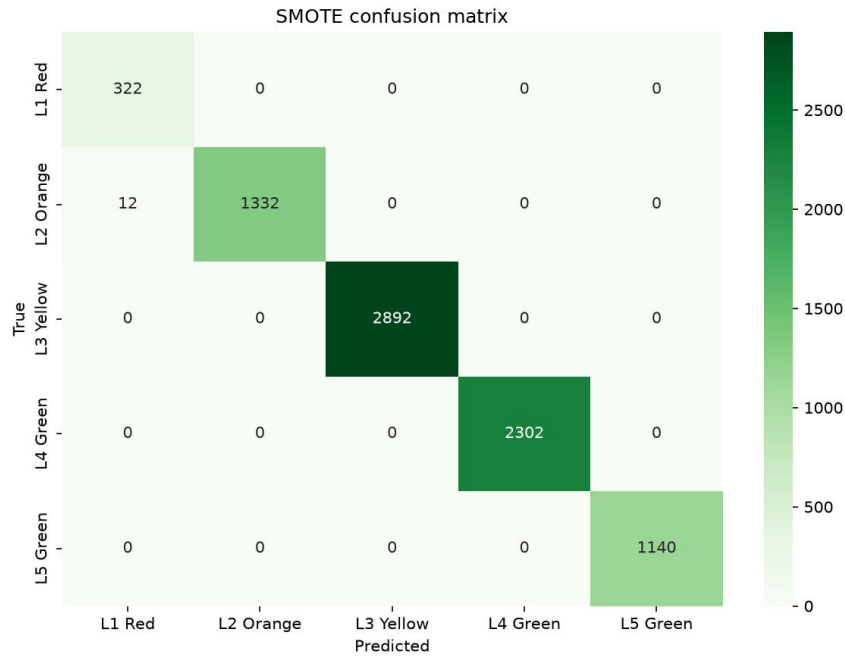


Figure 4.5: SMOTE-augmented BioBERT confusion matrix (same holdout split). Rows = true nurse label; columns = model prediction.

### Interpretation.

- **Chart type:** heatmap confusion matrix (same layout as Figure 4.4).
- **Axes:** rows = true nurse label; columns = model prediction; cell value = case count.
- **Data source:** SMOTE checkpoint, same holdout  $N = 8,000$ .
- **Key readings:** zero missed Red cases (no off-diagonal mass leaving the Red row); additional mass in the Red column from lower-urgency true labels (false urgent alerts).
- **Takeaway:** SMOTE trades improved Red sensitivity for a small number of false urgent alerts; nurse confirmation remains required (Berkowitz et al., 2019).

### 4.2.6 Per-class and headline comparison

Figure 4.6 compares per-class F1 scores. Both models exceed 0.98 on every level; the largest divergence is on Level 1, where SMOTE trades a small precision reduction for perfect recall.

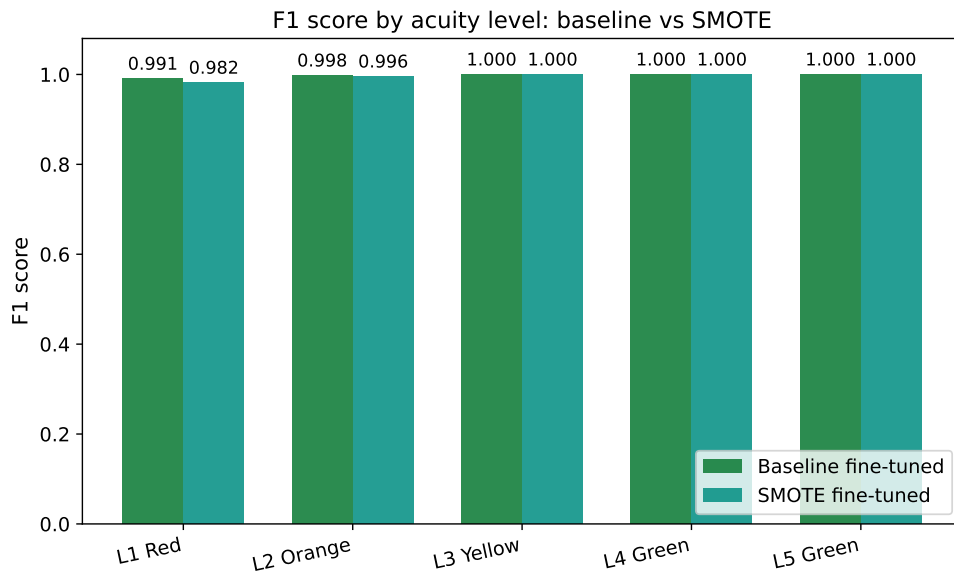


Figure 4.6: Per-class F1: baseline vs SMOTE on holdout data ( $N = 8,000$ ).

### Interpretation.

- **Chart type:** grouped bar chart.
- **Axes:** vertical axis = F1 score (0–1.05); horizontal axis = acuity level L1–L5; paired green/teal bars = baseline vs SMOTE.
- **Data source:** holdout evaluation, both checkpoints.
- **Key readings:** all levels exceed 0.98 F1 under both models; Level 1 shows the largest baseline–SMOTE gap.
- **Takeaway:** fine-tuning achieves balanced performance across all acuity levels, not only the majority Yellow class (Stewart et al., 2023).

Figure 4.7 summarises accuracy, macro-F1, and Level 1 recall side by side.

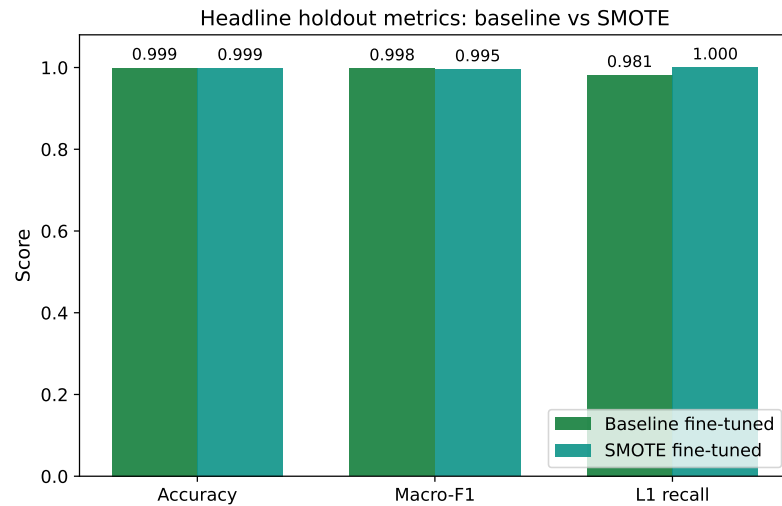


Figure 4.7: Headline holdout metrics: baseline vs SMOTE ( $N = 8,000$ ).

### Interpretation.

- **Chart type:** grouped bar chart.
- **Axes:** vertical axis = score (0–1.05); horizontal axis = metric name (accuracy, macro-F1, L1 recall); paired bars = baseline vs SMOTE.
- **Data source:** Table 4.1.
- **Key readings:** accuracy and macro-F1 differ by less than 0.004; L1 recall rises from 0.981 (baseline) to 1.000 (SMOTE).
- **Takeaway:** choose baseline for maximum Red precision; choose SMOTE when zero missed Red cases is the overriding safety requirement (Berkowitz et al., 2019).

Figure 4.8 compares recall for each acuity level under both models.

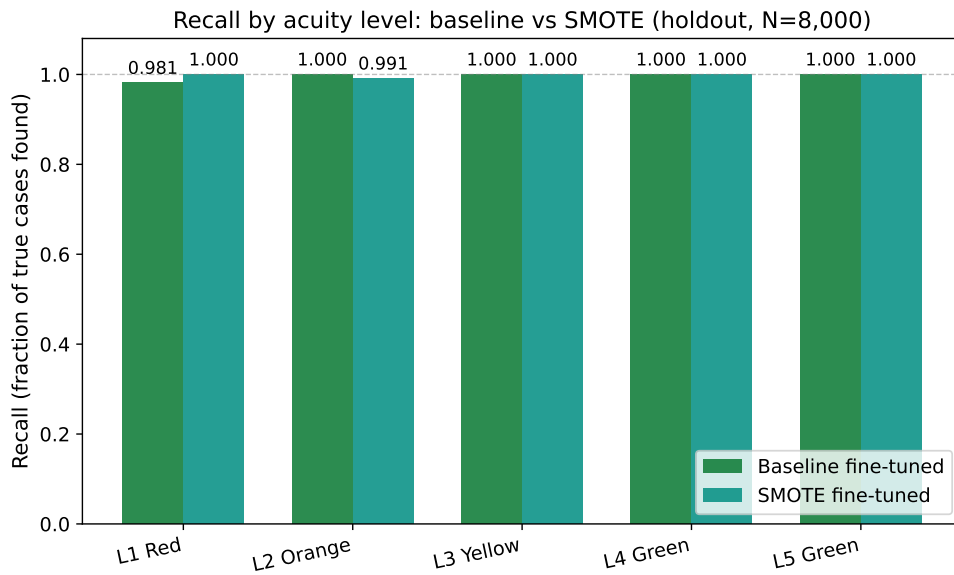


Figure 4.8: Recall by acuity level: baseline vs SMOTE (holdout,  $N = 8,000$ ).

### Interpretation.

- **Chart type:** grouped bar chart.
- **Axes:** vertical axis = recall, fraction of true cases correctly identified (0–1.05); horizontal axis = acuity level L1–L5; paired bars = baseline vs SMOTE.
- **Data source:** holdout per-class recall from `comparison.json`.
- **Key readings:** L1 recall 0.981 (baseline) vs 1.000 (SMOTE); L2–L5 remain near 1.0 under both models.
- **Takeaway:** Stage 3 fine-tuning overcomes much of the 9:1 class imbalance risk for urgent levels on in-distribution data (Chawla et al., 2002; Berkowitz et al., 2019).

Figure 4.9 compares inference latency statistics across both models.

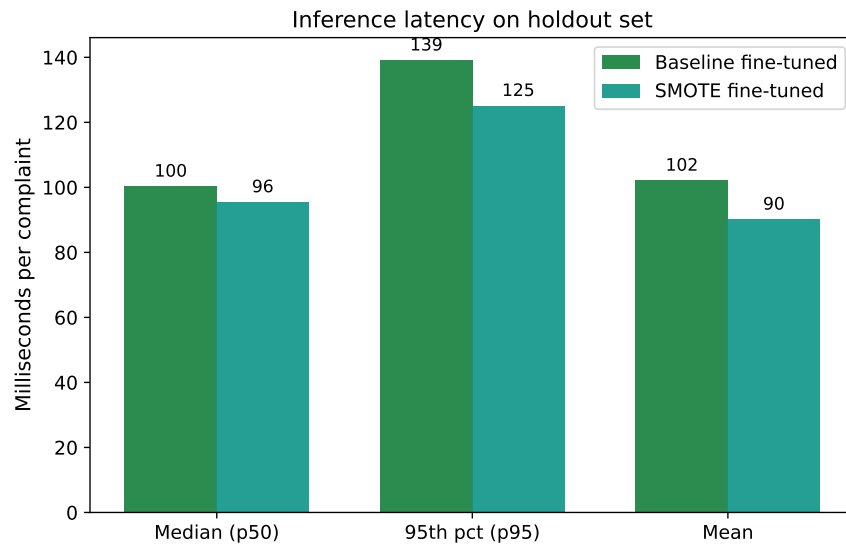


Figure 4.9: Inference latency on the 8,000-row holdout set (milliseconds per sample).

### Interpretation.

- **Chart type:** grouped bar chart.
- **Axes:** vertical axis = latency in milliseconds; horizontal axis = statistic (median p50, 95th percentile p95, mean); paired bars = baseline vs SMOTE.
- **Data source:** measured on evaluation hardware, holdout  $N = 8,000$ .
- **Key readings:** baseline p50 = 100.30 ms, p95 = 139.11 ms; SMOTE p50 = 95.50 ms.
- **Takeaway:** both checkpoints respond in well under one second, supporting real-time chatbot intake before nurse vitals are recorded (Stewart et al., 2023).

**Remark 4.1.** *High holdout accuracy on historical nurse labels indicates that chief complaints carry sufficient signal for acuity estimation before vitals are recorded (Stewart et al., 2023; Gligorijević et al., 2018). This measures label agreement, not live clinical outcomes (Section 4.2.2). Borderline or low-confidence predictions should trigger mandatory nurse review in deployment (South African Triage Group, 2012; Berkowitz et al., 2019).*

## 4.3 Inference latency

Real-time intake requires sub-second response (Stewart et al., 2023). On the evaluation hardware, baseline BioBERT achieves median latency = 100.30 ms per complaint (p95

= 139.11 ms; mean = 102.13 ms). The SMOTE checkpoint is slightly faster (p50 = 95.50 ms; mean = 90.17 ms) on the same split; see Figure 4.9.

At approximately 100 ms per complaint, the chatbot can return acuity probabilities, SATS colour, and a pathway card before a nurse completes manual registration. The clinical relevance gate (Section 3.17) rejects non-clinical input before BioBERT runs, saving compute and keeping triage staff focused on valid chief complaints. The prototype /predict API endpoint implements the same pipeline documented in Section 3.19.

## **4.4 Pathway routing at KNUST University Hospital**

The complete five-scenario pathway map (gate reject plus Red, Orange, Yellow, Green) is specified in Tables 3.21 and 3.22 and illustrated in Figure 3.10. Each scenario assigns primary clinical departments, support actions across Nursing, Pharmacy, Health Information, Hospital Information Technology, Administration, and Accounts, and an escalation rule (South African Triage Group, 2012; Rominski et al., 2014).

### **4.4.1 Nurse triage vs chatbot response**

Table 4.2 contrasts manual nurse triage with the hospital chatbot on dimensions relevant to intake at KNUST University Hospital. Nurse-side timings are drawn from published LMIC triage literature; chatbot timings are measured on the holdout set.

| Dimension                         | Manual nurse triage                                                                                                                                             | Hospital chatbot (this project)                                                                             |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| Time to first acuity signal       | Queue-dependent; LMIC intake often delayed minutes to hours before nurse contact (KATH Emergency Department, 2018; Mitchell et al., 2023; Nurchis et al., 2025) | Approximately 100 ms inference on evaluation hardware (Table 4.1, Figure 4.9)                               |
| Inputs used                       | Chief complaint + vitals + TEWS + clinical judgment (South African Triage Group, 2012)                                                                          | Text-only chief complaint; TEWS applied separately by nurse (Section 3.3)                                   |
| Output                            | SATS colour + bedside assessment                                                                                                                                | SATS colour + pathway card + confidence score                                                               |
| Validation basis                  | Live clinical decision at triage desk                                                                                                                           | Holdout agreement with historical nurse labels (Section 4.2.2)                                              |
| Role at KNUST University Hospital | Authoritative triage                                                                                                                                            | Triage assist at arrival; nurse override mandatory for low confidence and all vitals (Stewart et al., 2023) |

Table 4.2: Comparison of manual nurse triage and chatbot-assisted intake. Nurse timings are literature-based; chatbot latency is measured.

Once the chatbot assigns a colour in approximately 100 ms, pathway timer rules take effect: Orange cases require clinician contact within ten minutes, Yellow within 60 minutes, and Green within four hours (Section 3.16, Table 3.21) (South African Triage Group, 2012; Sundström et al., 2014). The chatbot does not replace nurse assessment; it provides an immediate acuity signal so that queue time before *any* routing decision is reduced relative to waiting for a free triage nurse (Mitchell et al., 2023; Nurchis et al., 2025).

**Gate rejection.** Non-clinical input receives no colour; Nursing re-prompts for a chief complaint. This prevents meaningless acuity scores and keeps acute queues free of idle interactions.

**Red.** Life-threatening text (e.g. crushing chest pain with dyspnoea) routes immediately to Surgery and Internal Medicine with Code Red alerts via HIT. Administration bypasses routine registration so resuscitation is not delayed (South African Triage Group, 2012).

**Orange.** Very urgent complaints (e.g. severe abdominal pain in pregnancy) activate Internal Medicine, Surgery, or Obstetrics and Gynaecology within ten minutes, with stat Diagnostics and a head-nurse escalation if unclaimed.

**Yellow.** For the running illustrative complaint (*“I have a headache and feel feverish. My body aches and I feel weak.”*), the model assigns level 3 at approximately 72% confidence. The pathway card routes to Outpatient urgent stream, Pediatrics



(when paediatric protocol applies), and parallel Diagnostics; Nursing sets a 60-minute reassessment timer.

**Green.** Routine presentations (e.g. dental pain without fever) divert to Outpatient, Dental, Eye, or Public and Preventive Health within four hours, reducing congestion in acute waiting areas (Yilmaz, 2021; Sundström et al., 2014).

Classification output (colour + confidence + department list) is the machine-readable input to this routing layer. Holdout evaluation (Section 4.2.2) bounds the reliability of automated pathway cards at intake on in-distribution historical text.

## 4.5 Anticipated impact on healthcare delivery

The following analysis interprets how the chatbot can support faster, clearer hospital flow at KNUST University Hospital, drawing on holdout metrics where available and on published triage literature elsewhere. No prospective waiting-time trial has yet been conducted at KNUST University Hospital (Mitchell et al., 2023).

### 4.5.1 Functional chatbot and workflow improvement

We anticipate a functional chatbot that accurately triages patients from a single chief complaint and improves hospital workflow by assigning SATS colour and department direction at intake (Stewart et al., 2023; Mitchell et al., 2023). Holdout evaluation reports 99.92% accuracy and Level 1 recall of 98.14% (or 100% under SMOTE), indicating that text-only classification aligns closely with nurse labels on historical in-distribution data (Section 4.2.2). Median inference latency of approximately 100 ms on evaluation hardware supports immediate digital response at arrival (Table 4.2).

Figure 4.10 compares time-to-first-acuity-signal: traditional intake (top) waits for a nurse; chatbot-assisted intake (bottom) classifies text immediately and runs nurse vitals in parallel.

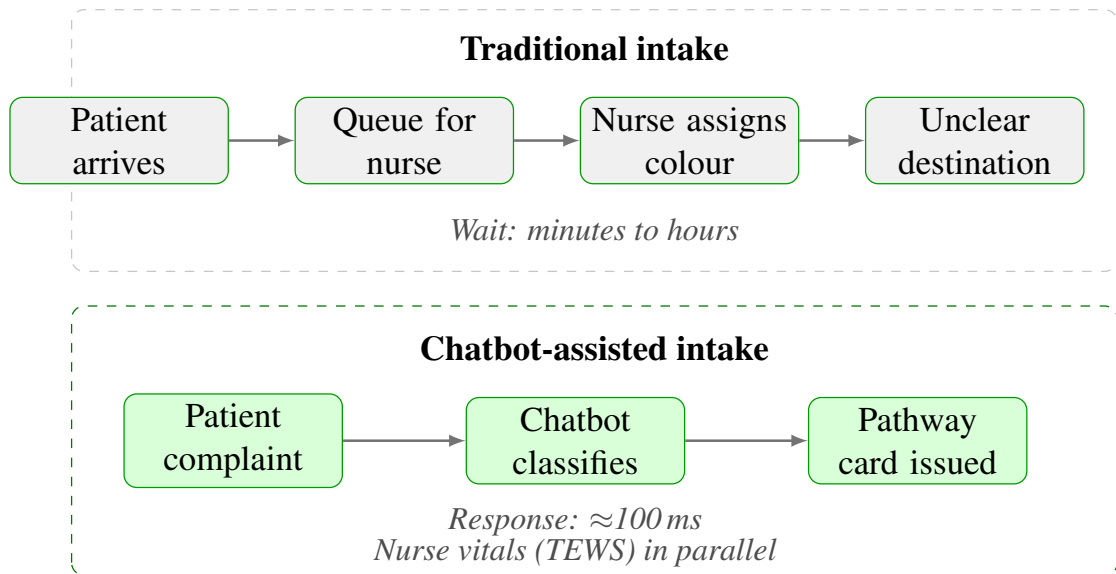


Figure 4.10: Stacked intake comparison at KNUST University Hospital. **Top:** patient queues minutes to hours before a nurse assigns colour. **Bottom:** chatbot classifies the complaint in  $\approx 100$  ms and issues a pathway card; nurse vitals continue in parallel. Colour routing (Red/Orange vs Green) is detailed in Section 4.4.

### Interpretation.

- **Chart type:** stacked workflow diagram (two panels, top and bottom).
- **Layout:** each panel is one horizontal chain read left to right. Top: arrive → queue → nurse assigns colour → unclear destination. Bottom: complaint → chatbot classifies → pathway card issued.
- **Data source:** pathway design (Chapter 3); chatbot latency from holdout evaluation (Figure 4.9); traditional wait times from LMIC triage literature (KATH Emergency Department, 2018; Mitchell et al., 2023).
- **Key readings:** top footnote “Wait: minutes to hours”; bottom footnotes “Response:  $\approx 100$  ms” and “Nurse vitals (TEWS) in parallel”. Red/Orange fast-track and Green digital queue are not shown here; see Section 4.4.
- **Takeaway:** the figure contrasts slow nurse-queue intake with immediate chatbot classification; nurse assessment is not replaced (South African Triage Group, 2012; Stewart et al., 2023).

### 4.5.2 Reduced initial triage wait times

Published triage literature suggests wait times for initial triage may decrease when patients interact with the bot immediately rather than queue for a nurse (Mitchell et al., 2023; Nurchis et al., 2025). Holdout evaluation shows the model assigns colour and

pathway in approximately 100 ms once valid clinical text is submitted (Table 4.2). Non-urgent (Green) cases can be digitally queued for later consultation while urgent cases are fast-tracked through Red and Orange escalation rules (Table 3.21), potentially reducing overall waiting duration (Sundström et al., 2014; Yilmaz, 2021). Parallel diagnostics orders flagged on Yellow and Green pathway cards may further shorten total time to definitive care (Sundström et al., 2014).

### **4.5.3 No overlooked critical cases**

The chatbot algorithm, combined with Red/Orange pathway escalation, is designed to trigger immediate alerts to hospital staff when language suggests life-threatening conditions (Gligorijević et al., 2018; South African Triage Group, 2012). Level 1 recall of 98.14% (baseline) or 100% (SMOTE) on holdout data supports this design intent on in-distribution text; the six baseline false negatives warrant nurse override and prospective audit (Berkowitz et al., 2019). Waiting-time deterioration increases adverse outcomes when triage colour is not reassessed (Nurchis et al., 2025); digital timers on Orange and Yellow pathway cards mitigate this risk (South African Triage Group, 2012).

### **4.5.4 Patient satisfaction and appropriate categorisation**

Patients may experience improved satisfaction when they receive quick feedback on urgency and feel guided through the system rather than left without direction (Stewart et al., 2023). Over time, the percentage of patients appropriately categorised (as confirmed by clinicians after assessment) will measure chatbot accuracy in live use. Holdout agreement with nurse labels (99.92% accuracy) is a necessary but not sufficient precondition for clinical confirmation rates (Section 4.2.2).

### **4.5.5 Staff workload and public-health insights**

Nursing staff may report reduced triage documentation burden, devoting more time to direct patient care and vitals (TEWS) rather than initial information gathering (Stewart et al., 2023; Mitchell et al., 2023). The gate rejects non-clinical messages before BioBERT, filtering idle interactions (Section 3.17). If scaled, anonymised complaint records could reveal public-health trends (e.g. common presentations by season), supporting Objective 4 in Chapter 1; privacy and governance review would be required before aggregation.

### 4.5.6 Success criteria

Project success would be measured by smoother patient flow, colour-coded pathways guiding patients to the right place at the right time, and a reduction in adverse events or delays attributable to triage (Rominski et al., 2014; KATH Emergency Department, 2018). Published NLP triage systems elsewhere report streamlined processes and optimised resource use in busy hospitals (Stewart et al., 2023; Mitchell et al., 2023).

**Limitations.** Holdout metrics measure classification accuracy on historical chief complaints, not patient outcomes, waiting-time reduction, or satisfaction scores (Mitchell et al., 2023; Nurchis et al., 2025). Prospective validation at KNUST University Hospital is required to quantify minutes saved and to confirm safe nurse override rates. Training data class imbalance may under-weight rare urgent phrasing; deployment should treat low-confidence or contradictory presentations as mandatory nurse review (Berkowitz et al., 2019; South African Triage Group, 2012). SMOTE increases Red recall but introduces false Red alerts (precision 96.41%); the baseline checkpoint is the project default unless clinical stakeholders prioritise zero missed Red cases. Holdout accuracy on the 80k corpus is an upper bound on offline label agreement, not a guarantee at live intake (Section 4.2.2).

## 4.6 Discussion of findings

Taken together, the model-first analysis and the holdout tables support three claims that answer the research questions in Chapter 1.

**Stage 3 is indispensable.** Figure 4.2 shows that biomedical pre-training and encoder transfer alone behave like chance or majority-class guessing for five-class acuity. Only after fine-tuning on nurse-labeled chief complaints does the Softmax head become a usable triage assist. That finding justifies treating Stage 3 as the project’s central machine-learning contribution, not as a minor calibration step.

**Colour follows acuity under  $f_{\text{SATS}}$ .** Because  $f_{\text{SATS}}$  is deterministic, colour-routing accuracy equals acuity accuracy except for the intentional collapse of levels 4 and 5 into Green. Operational mistakes therefore arise from wrong  $\hat{c}$ , not from an ambiguous colour map. Softmax confidence still matters: a Yellow prediction at 69% with substantial Green mass (Section 3.13) should not be treated as equivalent to a near-certain Red.

**Pathways make colour operational.** Tables 3.21–3.23 and Figure 4.10 show how a millisecond-scale text colour becomes a hospital action: timers, destinations, and

support departments. The anticipated impact section is necessarily interpretive until prospective time-motion data exist; the contribution at this stage is a complete, examinable protocol binding rather than a measured minutes-saved claim.

**Safety posture.** High holdout accuracy does not license unsupervised discharge decisions. The deployed posture is triage assist with nurse override, TEWS in parallel, gate rejection of non-clinical text, and mandatory review for low Softmax confidence or text–vitals conflict (Stewart et al., 2023; Berkowitz et al., 2019). That posture is consistent with SATS training manuals and with under-triage quality literature.

## CHAPTER 5

### CONCLUSION

#### 5.1 Summary

This project developed a mathematical pipeline from unstructured chief complaint text to South African Triage Scale (SATS) colour and clinical pathway at hospital intake. The chatbot accepts a single English free-text submission and returns acuity probabilities, a colour, and a pathway card aligned with SATS and KNUST University Hospital department structure.

Methodologically, the work specifies tokenisation  $\tau$ , contextual embeddings (word, position, and segment), twelve-layer BioBERT encoding with multi-head self-attention, Softmax classification as a Categorical distribution over five mutually exclusive acuity levels, the deterministic map  $f_{\text{SATS}}$ , a clinical relevance gate, and department routing across seventeen hospital units. TEWS is documented as the parallel nurse-facing vital-sign standard, not as a missing chatbot feature. Training is documented as a three-stage stack: PubMed masked language modelling (Stage 1), encoder transfer from a prior triage checkpoint (Stage 2), and comprehensive project fine-tuning on the Triagegeist English benchmark (80,000 pairs, Section 3.1.2) with a reinitialised five-class head (Stage 3).

Holdout evaluation reports 99.92% accuracy, 0.9977 macro-F1, and 98.14% Level 1 recall on 8,000 validation complaints, with approximately 100 ms inference latency (Chapter 4, Figures 4.9–4.7). A full five-scenario pathway map (gate reject, Red, Orange, Yellow, Green) routes each NLP outcome to clinical and support departments at KNUST University Hospital, addressing intake delay, queue congestion, and acuity discontinuity described in Chapter 1.

#### 5.2 Answers to research questions

- **RQ1–RQ2.** Yes on in-distribution Triagegeist English text: holdout acuity accuracy is 99.92%, and colour follows immediately from fixed  $f_{\text{SATS}}$  once  $\hat{c}$  is known.
- **RQ3.** Pathway cards  $P(C)$  are specified as  $(T_{\text{max}}, \text{destination}, \text{actions}, \text{escalation})$  with complete scenario tables and a seventeen-department routing map (Section 3.16).

- **RQ4.** The `Curatio /predict` prototype returns colour and pathway from text alone in about 100 ms; nurse TEWS continues in parallel when vitals exist.
- **RQ5.** The gate with threshold 0.35 rejects greeting-only and non-clinical input before BioBERT, as illustrated in Section 3.17.
- **RQ6.** Level 1 is the minority class; baseline Red recall is 98.14%, and SMOTE raises it to 100% at a small cost in overall accuracy and Red precision (Chapter 4).
- **RQ7.** Softmax confidence should trigger nurse oversight when the mode is not dominant (near ties) or when language conflicts with vitals; Softmax is a Categorical assist, not a clinical certificate.
- **RQ8.** Green pathways divert routine presentations to outpatient and specialty streams within four hours, reducing acute congestion while preserving nurse override.

## 5.3 Contributions restated

1. An examinable text-to-SATS-colour mathematical pipeline using BioBERT and Categorical Softmax.
2. Measured Stage 3 fine-tuning results (baseline and SMOTE) with latency suitable for intake assist.
3. Protocol-mapped colour-coded clinical pathways for KNUST University Hospital, including Green diversion and gate rejection.
4. A deployable chatbot API with medical gating, documented alongside TEWS as the vital-sign parallel path.

## 5.4 Limitations and future work

**Data scope.** Reported holdout metrics come exclusively from the Triagegeist synthetic English benchmark (Section 3.1.2); the model is English-only at intake. A site visit to KNUST University Hospital informed pathway design and generated supplementary English example complaints for prototype and qualitative analysis, but those examples were not used for fine-tuning. A natural next step is prospective collection of nurse-labeled chief complaints at KNUST University Hospital for local retraining, and exploring optional fusion of vitals and TEWS alongside free text when measurements become available.

**Class imbalance.** Chapter 4 already evaluates baseline and SMOTE-augmented checkpoints with stratified holdout metrics. Future work can refine minority-class sensitivity for urgent levels while preserving specificity on routine presentations, building on the imbalance-mitigation pipeline documented in Section 4.2.

**Clinical validation.** Holdout evaluation establishes strong offline agreement with nurse labels. The next phase is a prospective pilot at KNUST University Hospital to measure waiting times, nurse override rates, and patient outcomes in live intake, converting offline performance into operational evidence (Mitchell et al., 2023).

**Pathway refinement.** The five-scenario pathway map is SATS-aligned and complete for gate reject, Red, Orange, Yellow, and Green routing. Ongoing collaboration with hospital administrators and clinical leads at KNUST University Hospital will keep department routing tables current as service lines evolve (Rominski et al., 2014; South African Triage Group, 2012).

**Language and equity.** Extension to Ghanaian informal English and optional Twi-assisted intake would reduce exclusion of patients who do not write formal English complaints.

**System integration.** The prototype /predict API and pathway cards already anchor the chatbot in a deployable pipeline. Deeper integration with Hospital Information Technology and Health Information systems would automate timer alerts and bed assignment, completing the Flow Management Engine vision across Administration, Diagnostics, and all clinical departments listed in Table 3.23.



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## APPENDIX

### 5.5 Software repository

The implemented prototype and fine-tuned BioBERT artefacts are maintained at:

<https://github.com/samuel6814/triage>

No full source code is reproduced in this thesis; the mathematical specification in Chapter 3 and the system architecture in Figure 3.12 are the primary references.

### 5.6 Prototype /predict response fields

A successful clinical submission returns a JSON object whose fields mirror the mathematics of Chapter 3. Table 5.1 lists the principal keys.

| Field           | Meaning                                                                   |
|-----------------|---------------------------------------------------------------------------|
| gate_score      | Clinical relevance $g(X)$ in $[0, 1]$                                     |
| gate_passed     | True if $g(X) \geq 0.35$                                                  |
| probabilities   | Softmax vector $\hat{\mathbf{y}}$ over levels 1–5                         |
| predicted_level | $\hat{c} = \arg \max_c \hat{y}_c$                                         |
| sats_colour     | $C_{\text{NLP}} = f_{\text{SATS}}(\hat{c})$                               |
| pathway         | Pathway card $P(C)$ : $T_{\text{max}}$ , destination, actions, escalation |
| confidence      | Mode probability $\hat{y}_{\hat{c}}$ for nurse display                    |

Table 5.1: Principal /predict response fields and their mathematical counterparts.

Gate failures omit colour and pathway fields and instead return a prompt requesting a clinical chief complaint.

### 5.7 Stage 3 fine-tuning checklist

For reproducibility, Stage 3 training used the following checklist (full values in Table 3.18):

1. Join `Triagegeist train.csv` and `chief_complaints.csv` on `patient_id`.
2. Keep only English `chief_complaint_raw` and `triage_acuity`; drop vitals and demographics.
3. Stratified 90/10 split with seed 42 ( $N_{\text{train}} \approx 72,000$ ,  $N_{\text{eval}} = 8,000$ ).

4. Load BioBERT encoder; discard any prior two-class head; initialise  $W \in \mathbb{R}^{5 \times 768}$ ,  $\mathbf{b} \in \mathbb{R}^5$ .
5. Train three epochs, batch size 16, learning rate  $2 \times 10^{-5}$ , weight decay 0.01, max length 128, FP16 enabled.
6. Select the checkpoint with best validation cross-entropy (0.001812 at epoch 3 for the baseline run).
7. Optional SMOTE branch: oversample minority urgent classes before the same training recipe; compare on the *same* holdout split.

## 5.8 Glossary of principal symbols

| Symbol                         | Definition                                     |
|--------------------------------|------------------------------------------------|
| $X$                            | English chief-complaint string                 |
| $\tau$                         | WordPiece tokenisation map                     |
| $M$                            | Sequence length (maximum 128)                  |
| $D$                            | Hidden dimension (768)                         |
| $H^{(\ell)}$                   | Hidden-state matrix after encoder layer $\ell$ |
| $h_{[\text{CLS}]}$             | Summary vector (row 0 of $H^{(12)}$ )          |
| $W, \mathbf{b}$                | Classification weight matrix and bias          |
| $\mathbf{z}, \hat{\mathbf{y}}$ | Logits and Softmax (Categorical) probabilities |
| $\hat{c}$                      | Predicted acuity level                         |
| $f_{\text{SATS}}$              | Level-to-colour map                            |
| $C_{\text{NLP}}$               | NLP-assigned SATS colour                       |
| $P(C)$                         | Pathway card for colour $C$                    |
| $g(X)$                         | Clinical relevance gate score                  |
| $T$                            | TEWS integer from vitals                       |

Table 5.2: Compact glossary of symbols used across Chapters 3–4.